

Math Camp: Linear Algebra

August 12, 2014

Vector

- A vector is an n -tuple

Vector

- A vector is an n -tuple
- Vectors can specify direction and length in space

Vector

- A vector is an n -tuple
- Vectors can specify direction and length in space
- Each element of a vector is denoted with a subscript indicating its place in the vector.

Vector

- A vector is an n -tuple
- Vectors can specify direction and length in space
- Each element of a vector is denoted with a subscript indicating its place in the vector.
- The dimension of a vector is the number of its elements.

Vector Addition and scalar multiplication

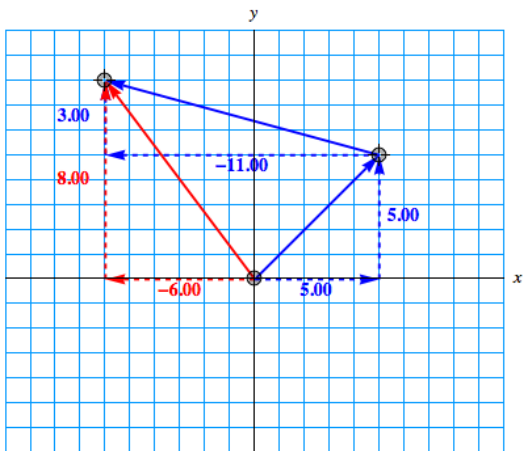
- $x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$

Vector Addition and scalar multiplication

- $x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $x - y = (x_1 - y_1, x_2 - y_2, \dots, x_n - y_n)$
- $\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n)$

Example of Vector Addition

Consider the vectors $x = (5, 5)$ and $y = (-11, 3)$, then $x + y = (-6, 8)$.



Metric Spaces: Euclidean Inner Product

Given $x, y \in \mathbb{R}^n$, the Euclidean inner product of the vectors x and y , denoted $x \cdot y$ is defined as:

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

Metric Spaces: Euclidean Inner Product

Given $x, y \in \mathbb{R}^n$, the Euclidean inner product of the vectors x and y , denoted $x \cdot y$ is defined as:

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

Example: $x, y \in \mathbb{R}^2$, $x = (1, 3)$, and $y = (2, 4)$. Then the dot product is $1 \cdot 2 + 3 \cdot 4 = 14$

Properties of the Inner Product

The inner product has the following properties for any vectors $x, y, z \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$

Properties of the Inner Product

The inner product has the following properties for any vectors $x, y, z \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$

- Symmetry: $x \cdot y = y \cdot x$

Properties of the Inner Product

The inner product has the following properties for any vectors $x, y, z \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$

- Symmetry: $x \cdot y = y \cdot x$
- Bilinearity: $(ax + by) \cdot z = ax \cdot z + by \cdot z$ and $x \cdot (ay + bz) = x \cdot ay + x \cdot bz$

Properties of the Inner Product

The inner product has the following properties for any vectors $x, y, z \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$

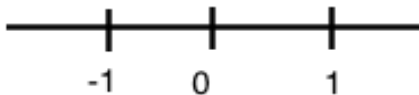
- Symmetry: $x \cdot y = y \cdot x$
- Bilinearity: $(ax + by) \cdot z = ax \cdot z + by \cdot z$ and $x \cdot (ay + bz) = x \cdot ay + x \cdot bz$
- Positivity: $x \cdot x \geq 0$ with equality holding iff $x = 0$

The Euclidean Norm in Euclidean Space

In Euclidean space the norm is the length of the vector or can be thought of as the distance from the origin.

The Euclidean Norm in Euclidean Space

In Euclidean space the norm is the length of the vector or can be thought of as the distance from the origin. For example consider $x = 1 \in \mathbb{R}^1$. $\|x\| = (1 \cdot 1)^{1/2} = 1$. Or consider $x = -1 \in \mathbb{R}^1$. $\|x\| = (-1 \cdot -1)^{1/2} = 1$



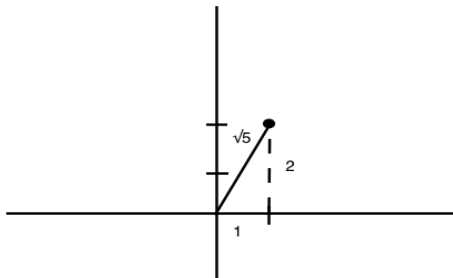
The Euclidean Norm in Euclidean Space

In Euclidean space the norm is the length of the vector or can be thought of as the distance from the origin. For example consider

$x = 1 \in \mathbb{R}^1$. $\|x\| = (1 \cdot 1)^{1/2} = 1$. Or consider $x = -1 \in \mathbb{R}^1$.

$\|x\| = (-1 \cdot -1)^{1/2} = 1$ As another example, consider

$x = (1, 2) \in \mathbb{R}^2$. $\|x\| = (1^2 + 2^2)^{1/2} = 5^{1/2}$



The Euclidean Norm in Euclidean Space

In Euclidean space the norm is the length of the vector or can be thought of as the distance from the origin. For example consider $x = 1 \in \mathbb{R}^1$. $\|x\| = (1 \cdot 1)^{1/2} = 1$. Or consider $x = -1 \in \mathbb{R}^1$. $\|x\| = (-1 \cdot -1)^{1/2} = 1$ As another example, consider $x = (1, 2) \in \mathbb{R}^2$. $\|x\| = (1^2 + 2^2)^{1/2} = 5^{1/2}$

$$\text{Generally } \|x\| = \sqrt{(x_1^2 + x_2^2 + \dots + x_n^2)}$$

The Taxicab Norm

Although we don't make much use of this norm, another norm that is used to define Euclidean distance is the taxicab norm

$$\|x\|_1 = \sum_{i=1}^n (x_i^2)^{1/2}$$

The Taxicab Norm

Although we don't make much use of this norm, another norm that is used to define Euclidean distance is the taxicab norm

$$\|x\|_1 = \sum_{i=1}^n (x_i^2)^{1/2}$$

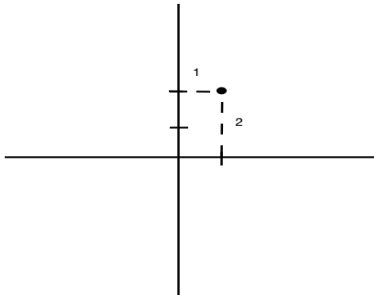
Remember the euclidean norm is $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$.

The Taxicab Norm

Although we don't make much use of this norm, another norm that is used to define Euclidean distance is the taxicab norm

$$\|x\|_1 = \sum_{i=1}^n (x_i^2)^{1/2}$$

Remember the euclidean norm is $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$. The taxicab norm defines the block distance from the origin.



Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$
 - A point either has a positive or no distance from the origin

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$
 - A point either has a positive or no distance from the origin
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$
 - A point either has a positive or no distance from the origin
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$
 - When one multiplies a point by a constant scalar, the resulting point will be the distance of the original point from the origin multiplied by the magnitude of the scalar

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$
 - A point either has a positive or no distance from the origin
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$
 - When one multiplies a point by a constant scalar, the resulting point will be the distance of the original point from the origin multiplied by the magnitude of the scalar
- Triangle Inequality: $\|x + y\| \leq \|x\| + \|y\|$

Properties of the Euclidean Norm

The norm satisfies the the following properties at all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$

- Positivity: $\|x\| \geq 0$ with equality iff $x = 0$
 - A point either has a positive or no distance from the origin
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$
 - When one multiplies a point by a constant scalar, the resulting point will be the distance of the original point from the origin multiplied by the magnitude of the scalar
- Triangle Inequality: $\|x + y\| \leq \|x\| + \|y\|$
 - The shortest distance from the origin to a point is a straight line.

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

- Positivity: $\|x\| \geq 0$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

■ Positivity: $\|x\| \geq 0$

□ $\|x\| = (2^2 + 0)^{1/2} = 2$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

- Positivity: $\|x\| \geq 0$
 - $\|x\| = (2^2 + 0)^{1/2} = 2$
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

■ Positivity: $\|x\| \geq 0$

□ $\|x\| = (2^2 + 0)^{1/2} = 2$

■ Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$

□ $\|\alpha x\| = ((\alpha 2)^2 + (\alpha 0)^2)^{1/2} = \alpha(2^2 + 0^2)^{1/2} = \alpha 2$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

■ Positivity: $\|x\| \geq 0$

□ $\|x\| = (2^2 + 0)^{1/2} = 2$

■ Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$

□ $\|\alpha x\| = ((\alpha 2)^2 + (\alpha 0)^2)^{1/2} = \alpha(2^2 + 0^2)^{1/2} = \alpha 2$

■ Triangle Inequality: $\|x + y\| \leq \|x\| + \|y\|$

Properties of Euclidean Norm: Example

To provide some intuition of the previous properties consider an example in $\mathbb{R}^2$, where $x = (2, 0)$ and $y = (1, 1)$. Remember the definition of the norm is: $\|x\| = (\sum_{i=1}^n x_i^2)^{1/2}$

■ Positivity: $\|x\| \geq 0$

$$\square \|x\| = (2^2 + 0)^{1/2} = 2$$

■ Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$

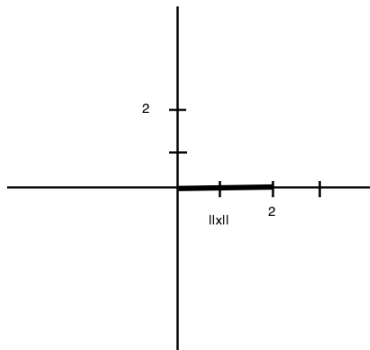
$$\square \|\alpha x\| = ((\alpha 2)^2 + (\alpha 0)^2)^{1/2} = \alpha(2^2 + 0^2)^{1/2} = \alpha 2$$

■ Triangle Inequality: $\|x + y\| \leq \|x\| + \|y\|$

$$\square \|x + y\| = ((1 + 2)^2 + (1 + 0)^2)^{1/2} = 10^{1/2} < (2^2 + 0)^{1/2} + (1^2 + 1^2)^{1/2} = 2 + 2^{1/2} = \|x\| + \|y\|$$

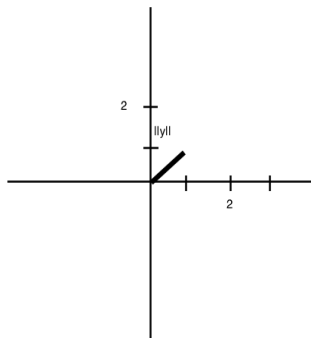
Visualizing Properties of Euclidean Norm: Positivity

Positivity: $\|x\| \geq 0$



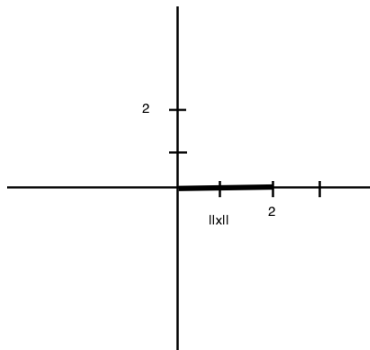
Visualizing Properties of Euclidean Norm: Positivity

Positivity: $\|x\| \geq 0$



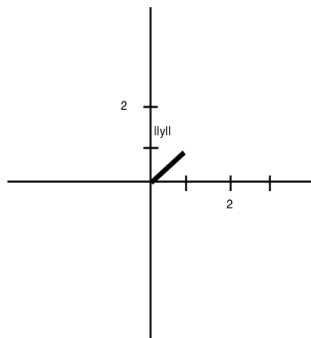
Visualizing Properties of Euclidean Norm: Positivity

Positivity: $\|x\| \geq 0$



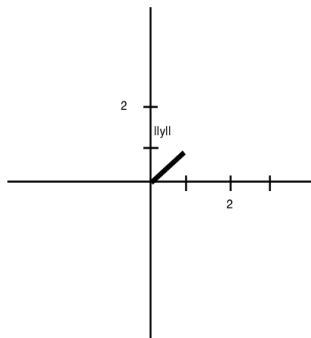
Visualizing Properties of Euclidean Norm: Positivity

Positivity: $\|x\| \geq 0$



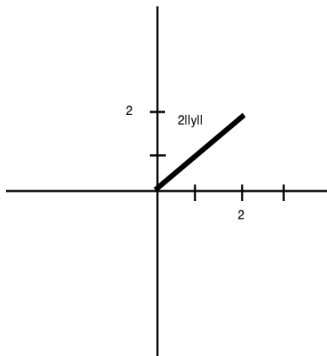
Visualizing Properties of Euclidean Norm: Homogeneity

Homogeneity: $\|\alpha y\| = |\alpha| \cdot \|y\|$ (If $\alpha = 2$ and $y = (1, 1)$ then $\|y\| = \sqrt{2}$, $\|\alpha y\| = \alpha\|y\| = 2\sqrt{2}$.)



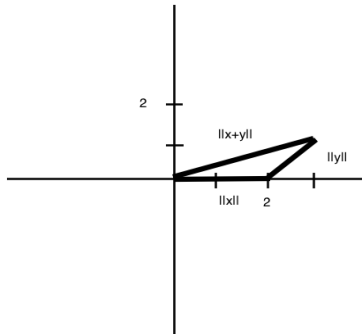
Visualizing Properties of Euclidean Norm: Homogeneity

Homogeneity: $\|\alpha y\| = |\alpha| \cdot \|y\|$ (If $\alpha = 2$ and $y = (1, 1)$ then $\|y\| = \sqrt{2}$, $\|\alpha y\| = \alpha\|y\| = 2\sqrt{2}$.)



Visualizing Properties of Euclidean Norm: Triangle Inequality

Triangle Inequality: $\|x + y\| \leq \|x\| + \|y\|$



Euclidean Distance

The *Euclidean Distance* $d(x, y)$ between two vectors x and y in $\mathbb{R}^n$ is given by:

$$d(x, y) = (\sum_{i=1}^n (x_i - y_i)^2)^{1/2}$$

The distance function d is called a *metric* and is related to the norm $||\cdot||$ through the identity

$$d(x, y) = ||x - y|| \quad \forall x, y \in \mathbb{R}^n$$

Euclidean Distance is a generalization of the Euclidean Norm.

Systems of Linear Equations

Linear Algebra is the study of linear equations. A system of m linear equations in n variables has the the form

$$\begin{aligned}b_1 &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\b_2 &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\&\vdots \\b_m &= a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n\end{aligned}$$

Systems of Linear Equations

The linear system can also be written in matrix form.

$$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

We can write the linear system as $\mathbf{b} = A\mathbf{X}$, where A is an $m \times n$ matrix, $\mathbf{b}$ is an $m \times 1$ vector, and $\mathbf{x}$ is an $n \times 1$ vector. To study linear systems we study matrices and their properties. Finally notice that the elements in a matrix are numbered a_{ij} where i is the row and j is the column where the element is located.

Basic Matrix Operation

Consider two $n \times m$ matrices:

$$\mathbf{A} = \begin{pmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \vdots & \ddots \\ a_{n1} & \dots & a_{nm} \end{pmatrix}, \mathbf{B} = \begin{pmatrix} b_{11} & \dots & b_{1m} \\ \vdots & \vdots & \ddots \\ b_{n1} & \dots & b_{nm} \end{pmatrix}$$

The basic matrix operation are:

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} + b_{11} & \dots & a_{1m} + b_{1m} \\ \vdots & \vdots & \ddots \\ a_{n1} + b_{n1} & \dots & a_{nm} + b_{nm} \end{pmatrix},$$

$$\lambda \mathbf{A} = \begin{pmatrix} \lambda a_{11} & \dots & \lambda a_{1m} \\ \vdots & \vdots & \ddots \\ \lambda a_{n1} & \dots & \lambda a_{nm} \end{pmatrix}, \text{ where } \lambda \in \mathbb{R}$$

Matrix Multiplication

To multiply the matrices CD , the number of the columns in the C matrix must equal the number of rows in the D matrix. Say that C is an $n \times m$ matrix, and D is an $m \times k$ matrix. Then

$$\mathbf{E} = \mathbf{CD} = \begin{pmatrix} \sum_{q=1}^m c_{1,q} d_{q,1} & \cdots & \sum_{q=1}^m c_{1,q} d_{q,k} \\ \vdots & \vdots & \ddots \\ \sum_{q=1}^m c_{n,q} d_{q,1} & \cdots & \sum_{q=1}^m c_{n,q} d_{q,k} \end{pmatrix}$$

Dot Product or Inner Product

The inner product a scalar that results when two vectors of the same length are multiplied together.

$$v \cdot z = vz' = (v_1 \dots v_n) \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} = \sum_{i=1}^n v_i z_i$$

Dot Product or Inner Product

The inner product is a scalar that results when two vectors of the same length are multiplied together.

$$v \cdot z = vz' = (v_1 \dots v_n) \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} = \sum_{i=1}^n v_i z_i$$

Note that any element in a matrix E , that is the product of two matrices, CD is a dot product. For example, the first element in the matrix E is

$$(c_{11} \dots c_{1n}) \begin{pmatrix} d_{11} \\ \vdots \\ d_{n1} \end{pmatrix} = \sum_{q=1}^n c_{1q} d_{q1}$$

Matrix Multiplication Example

Consider the matrices $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$. What is AB ?

Matrix Multiplication Example

Consider the matrices $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$. What is

AB ? $AB = \begin{pmatrix} 1(3) + 4(2) & 1(2) + 4(3) \\ 2(3) + 1(2) & 2(2) + 1(3) \end{pmatrix} = \begin{pmatrix} 11 & 14 \\ 8 & 7 \end{pmatrix}$. What is

BA ?

Matrix Multiplication Example

Consider the matrices $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$. What is

AB ? $AB = \begin{pmatrix} 1(3) + 4(2) & 1(2) + 4(3) \\ 2(3) + 1(2) & 2(2) + 1(3) \end{pmatrix} = \begin{pmatrix} 11 & 14 \\ 8 & 7 \end{pmatrix}$. What is

BA ? $BA = \begin{pmatrix} 7 & 14 \\ 8 & 11 \end{pmatrix}$

Matrix Multiplication Example 2

Consider the matrices $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. What is

AB ?

Matrix Multiplication Example 2

Consider the matrices $A = \begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. What is AB ? $AB = \begin{pmatrix} 11 \\ 8 \end{pmatrix}$. What is BA ?

Zero Matrix

A zero matrix is a matrix where each element is 0

$$\mathbf{0} = \begin{pmatrix} 0 & \dots & 0 \\ \vdots & \vdots & \ddots \\ 0 & \dots & 0 \end{pmatrix}$$

Identity Matrix

The identity matrix is a $n \times n$ matrix with zeros everywhere except along the diagonal. Note that the number of columns must equal the number of rows.

$$\mathbf{I} = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

Identity Matrix

The identity matrix is a $n \times n$ matrix with zeros everywhere except along the diagonal. Note that the number of columns must equal the number of rows.

$$\mathbf{I} = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

- If A and B are any $k \times n$ and $n \times m$ matrices, then $AI = A$ and $IB = B$

Identity Matrix

The identity matrix is a $n \times n$ matrix with zeros everywhere except along the diagonal. Note that the number of columns must equal the number of rows.

$$\mathbf{I} = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

- If A and B are any $k \times n$ and $n \times m$ matrices, then $AI = A$ and $IB = B$
- If C is an $n \times n$ matrix then $IC = CI = C$

Identity Matrix

The identity matrix is a $n \times n$ matrix with zeros everywhere except along the diagonal. Note that the number of columns must equal the number of rows.

$$\mathbf{I} = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$$

- If A and B are any $k \times n$ and $n \times m$ matrices, then $AI = A$ and $IB = B$
- If C is an $n \times n$ matrix then $IC = CI = C$
- Also $I^2 = I$.

Square, Symmetric, and Transpose Matrices

A *square matrix* is a matrix that has the same number of columns and rows. As noted the identity matrix must be square. If it wasn't then $AI \neq A$. If a square matrix has the property that $a_{i,j} = a_{j,i}$ for all of its elements, then we call it a symmetric matrix.

Square, Symmetric, and Transpose Matrices

A *square matrix* is a matrix that has the same number of columns and rows. As noted the identity matrix must be square. If it wasn't then $AI \neq A$. If a square matrix has the property that $a_{i,j} = a_{j,i}$ for all of its elements, then we call it a symmetric matrix. The transpose of a matrix A , denoted A' is a matrix such that for each element of A' , $a'_{i,j} = a_{j,i}$. For example the transpose of the matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \text{ is } \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}.$$

Note that a matrix A is symmetric if $A = A'$

Properties of the transpose

- $(A')' = A$
- $(A + B)' = A' + B'$
- $(\alpha A)' = \alpha A'$
- $(AB)' = B'A'$, which can be extended: $(ABC)' = C'B'A'$
- If the matrix A is $n \times k$, then A' is $k \times n$.

Diagonal Matrices

A square matrix A is diagonal if it is all zeros except along the diagonal:

$$\begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

All diagonal matrices are also symmetric.

Triangular Matrices

A square matrix A is upper triangular if all of its entries below the diagonal are zero

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

and is lower triangular if all its entries above the diagonal are zero

$$\begin{pmatrix} a_{11} & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse
 - Suppose that B and C are both inverses of A . Then
$$C = CI = C(AB) = (CA)B = IB = B$$

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse
 - Suppose that B and C are both inverses of A . Then
$$C = CI = C(AB) = (CA)B = IB = B$$
- $(A^{-1})^{-1} = A$

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse
 - Suppose that B and C are both inverses of A . Then
$$C = CI = C(AB) = (CA)B = IB = B$$
- $(A^{-1})^{-1} = A$
- $(\alpha A)^{-1} = \frac{1}{\alpha}(A)^{-1}$

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse
 - Suppose that B and C are both inverses of A . Then
$$C = CI = C(AB) = (CA)B = IB = B$$
- $(A^{-1})^{-1} = A$
- $(\alpha A)^{-1} = \frac{1}{\alpha}(A)^{-1}$
- $(AB)^{-1} = B^{-1}A^{-1}$ if B^{-1}, A^{-1} exist.

Inverse Matrices

If there exists a matrix B such that $BA = AB = I$, then we call B the inverse of A and denote it A^{-1} . A can only have an inverse if it is a square matrix, but not all square matrices have an inverse.

- A square matrix A can have at most one inverse
 - Suppose that B and C are both inverses of A . Then
$$C = CI = C(AB) = (CA)B = IB = B$$
- $(A^{-1})^{-1} = A$
- $(\alpha A)^{-1} = \frac{1}{\alpha}(A)^{-1}$
- $(AB)^{-1} = B^{-1}A^{-1}$ if B^{-1}, A^{-1} exist.
- $(A')^{-1} = (A^{-1})'$

Orthogonal and Idempotent Matrices

- A matrix A is orthogonal if $A'A = I$. So a matrix is orthogonal if $A^{-1} = A'$.

Orthogonal and Idempotent Matrices

- A matrix A is orthogonal if $A'A = I$. So a matrix is orthogonal if $A^{-1} = A'$.
- A matrix is idempotent if it is both symmetric and $AA = A$

Application of the Idempotent Matrix: OLS

Chose a vector B of coefficient estimates to minimize the sum of the squared residuals. In matrix form, minimize:

$$(Y - XB)'(Y - XB)$$

The resulting estimator is $B = (X'X)^{-1}X'Y$ The vector of residuals is

$$e = Y - XB = Y - (X'X)^{-1}X'Y = [I - X(X'X)^{-1}X']Y = MY$$

Application of the Idempotent Matrix: OLS

Chose a vector B of coefficient estimates to minimize the sum of the squared residuals. In matrix form, minimize:

$$(Y - XB)'(Y - XB)$$

The resulting estimator is $B = (X'X)^{-1}X'Y$ The vector of residuals is

$$e = Y - XB = Y - (X'X)^{-1}X'Y = [I - X(X'X)^{-1}X']Y = MY$$

Here both M and $X(X'X)^{-1}X'$ are idempotent, which allows the simplification when the sum of squared residuals are computed:

$$e'e = (MY)'(MY) = Y'M'MY = Y'MMY = Y'MY$$

Trace

The **trace** of an $n \times n$ square matrix is the sum of its diagonal elements.

$$\text{Tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

The Determinant: 1×1 matrix

For square matrices we can define a number for the matrices that is called the determinant. This number provides important information, such as telling us if a linear system of equations has a unique solution. First, we just present the algorithm used to compute it. The determinant can be defined inductively:

The Determinant: 1×1 matrix

For square matrices we can define a number for the matrices that is called the determinant. This number provides important information, such as telling us if a linear system of equations has a unique solution. First, we just present the algorithm used to compute it. The determinant can be defined inductively:

- The determinant of a 1×1 matrix (a) is a and is denoted $\det(a)$.

The Determinant: 2×2 matrix

The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$.

The Determinant: 2×2 matrix

The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$. This is the same as $a \cdot \det(d) - b \cdot \det(c)$. The first term is the $(1,1)$ th entry of A times the determinant of the submatrix that is obtained by deleting the first row and column from A (the row and column that contained the $(1,1)$ th entry.) Similarly the second term is obtained by multiplying the $(1,2)$ th entry by the determinant that is obtained by deleting the row and column that contains this entry. The signs alternate with the first term being added and the second term being subtracted.

The Determinant: 3×3 matrix

The determinant of a 3×3 matrix $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is

$$aei + bfg + cdh - ceg - bdi - afh.$$

This can also be written as:

$$a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

Can you see the pattern?

The Determinant: 3×3 matrix

The determinant of a 3×3 matrix $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is

$$aei + bfg + cdh - ceg - bdi - afh.$$

This can also be written as:

$$a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

Can you see the pattern? To obtain a determinant:

- Multiple each element in the top row with the determinant of the matrix left when we delete the row and column in which the respective element resides.

The Determinant: 3×3 matrix

The determinant of a 3×3 matrix $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ is

$$aei + bfg + cdh - ceg - bdi - afh.$$

This can also be written as:

$$a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

Can you see the pattern? To obtain a determinant:

- Multiple each element in the top row with the determinant of the matrix left when we delete the row and column in which the respective element resides.
- The signs of the terms alternate, starting with positive.

Minor and Cofactor

To define a determinant, we must first define a *minor* and *cofactor*.

Minor and Cofactor

To define a determinant, we must first define a *minor* and *cofactor*.

- Let A be an $n \times n$ matrix. Let $A_{i,j}$ be the $(n-1) \times (n-1)$ submatrix obtained by deleting row i and column j from A . Then the scalar $M_{i,j} = \det(A_{i,j})$ is called the (i,j) th *minor* of A .

Minor and Cofactor

To define a determinant, we must first define a *minor* and *cofactor*.

- Let A be an $n \times n$ matrix. Let $A_{i,j}$ be the $(n-1) \times (n-1)$ submatrix obtained by deleting row i and column j from A . Then the scalar $M_{i,j} = \det(A_{i,j})$ is called the (i,j) th *minor* of A .
- The scalar $C_{ij} = -1^{i+j} M_{i,j}$ is called the (i,j) th *cofactor* of A . The cofactor is the signed minor.

Minor and Cofactor

To define a determinant, we must first define a *minor* and *cofactor*.

- Let A be an $n \times n$ matrix. Let $A_{i,j}$ be the $(n-1) \times (n-1)$ submatrix obtained by deleting row i and column j from A . Then the scalar $M_{i,j} = \det(A_{i,j})$ is called the (i,j) th *minor* of A .
- The scalar $C_{ij} = -1^{i+j} M_{i,j}$ is called the (i,j) th *cofactor* of A . The cofactor is the signed minor.
 - Generally, the cofactor is negative when either i or j is even and the other is odd. The cofactor is even with both the i and j are even or odd.

Minor and Cofactor: Example

Consider the matrix A :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Minor and Cofactor: Example

Consider the matrix A :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The submatrix that is obtained by eliminating the 2nd row and the 2nd column is:

$$\mathbf{A}_{22} = \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}$$

What is the $(2, 2)$ th minor of A ?

Minor and Cofactor: Example

Consider the matrix A :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The submatrix that is obtained by eliminating the 2nd row and the 2nd column is:

$$\mathbf{A}_{22} = \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}$$

What is the $(2, 2)$ th minor of A ? $M_{22} = a_{11}a_{33} - a_{13}a_{31}$

Minor and Cofactor: Example

Consider the matrix A :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The submatrix that is obtained by eliminating the 2nd row and the 2nd column is:

$$\mathbf{A}_{22} = \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}$$

What is the $(2, 2)$ th minor of A ? $M_{22} = a_{11}a_{33} - a_{13}a_{31}$

What is the $(2, 2)$ th cofactor of A ?

Minor and Cofactor: Example

Consider the matrix A :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

The submatrix that is obtained by eliminating the 2nd row and the 2nd column is:

$$\mathbf{A}_{22} = \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix}$$

What is the $(2, 2)$ th minor of A ? $M_{22} = a_{11}a_{33} - a_{13}a_{31}$

What is the $(2, 2)$ th cofactor of A ?

$$C_{22} = (-1)^{2+2}M_{22} = a_{11}a_{33} - a_{13}a_{31}.$$

Defining the Determinant

Remember the determinant of the 2×2 matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, is defined as $a \cdot \det(d) - b \cdot \det(c)$. Using the minor and cofactor the determinant can also be defined as

$$\det(A) = aM_{11} - bM_{12} = aC_{11} + bC_{12}$$

The determinant of the 3×3 is

$$a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

This can also be defined as

$$\det(A) = aM_{11} - bM_{12} + cM_{13} = aC_{11} + bC_{12} + cC_{13}$$

General Definition of the Determinant

The determinant of an $n \times n$ matrix is defined as:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$$

Notice that the definition only uses the elements and cofactors for the top row of the matrix, which is called the *cofactor expansion along the first row*. But, a cofactor expansion along any row or column will be equal to the determinant.

Finding the Determinant: Example

Find the determinant of the upper diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$$

The determinant is

$$aC_{11}+bC_{12}+cC_{13} = a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

Finding the Determinant: Example

Find the determinant of the upper diagonal matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$$

The determinant is

$$\begin{aligned} aC_{11} + bC_{12} + cC_{13} &= a \cdot \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} + c \cdot \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} \\ &= 1 \cdot \det \begin{pmatrix} 3 & 0 \\ 5 & 6 \end{pmatrix} - 0 \cdot \det \begin{pmatrix} 2 & 0 \\ 4 & 6 \end{pmatrix} + 0 \cdot \det \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = 1 \cdot 3 \cdot 6 = 18 \end{aligned}$$

Finding the Determinant: Example Continued

Find the determinant of matrix using a cofactor expansion of the second column

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$$

The determinant is

$$0C_{12} + 3C_{22} + 5C_{32}$$

Finding the Determinant: Example Continued

Find the determinant of matrix using a cofactor expansion of the second column

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$$

The determinant is

$$0C_{12} + 3C_{22} + 5C_{32}$$

$$= -0 \cdot \det \begin{pmatrix} 2 & 0 \\ 4 & 6 \end{pmatrix} + 3 \cdot \det \begin{pmatrix} 1 & 0 \\ 4 & 6 \end{pmatrix} - 5 \cdot \det \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} = 3 \cdot 1 \cdot 6 = 18$$

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$
- If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$
- If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$
- $\det(A') = \det(A)$

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$
- If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$
- $\det(A') = \det(A)$
- **Definition** An $n \times n$ matrix A is called *singular* if $\det(A) = 0$. It is called *non-singular* if $\det(A) \neq 0$.

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$
- If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$
- $\det(A') = \det(A)$
- **Definition** An $n \times n$ matrix A is called *singular* if $\det(A) = 0$. It is called *non-singular* if $\det(A) \neq 0$.

Properties of the Determinant

- $\det(I) = 1$, where I is the identity matrix
- $\det(AB) = \det(A)\det(B)$
- If A is invertible, $\det(A^{-1}) = \frac{1}{\det(A)}$
- $\det(A') = \det(A)$
- **Definition** An $n \times n$ matrix A is called *singular* if $\det(A) = 0$. It is called *non-singular* if $\det(A) \neq 0$.
 - The inverse of a singular matrix does not exist

Adjoint Definition

Definition For any $n \times n$ matrix A , let C_{ij} denote the (i, j) th cofactor of A (-1^{i+j} times the determinant of the submatrix obtained by deleting row i and column j from A .) Then the $n \times n$ matrix whose (i, j) th entry is C_{ji} , the (j, i) th cofactor of A (note the switch in indices) is called the adjoint of A and is written $\text{adj}A$.

Adjoint Definition: Example

For example

$$A = \begin{pmatrix} 2 & 4 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Adjoint Definition: Example

For example

$$A = \begin{pmatrix} 2 & 4 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$C_{11} = \begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} = 3, C_{12} = - \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0, \dots$$

Adjoint Definition: Example

For example

$$A = \begin{pmatrix} 2 & 4 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$C_{11} = \begin{vmatrix} 3 & 0 \\ 0 & 1 \end{vmatrix} = 3, C_{12} = - \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0, \dots$$

$$\text{adj}A = \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$$

Let A be a nonsingular matrix. Then,

$$\blacksquare A^{-1} = \frac{1}{\det A} \cdot \text{adj}A$$

Solving Linear Systems

Recall that the system of equations

$$\begin{aligned}b_1 &= a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\b_2 &= a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\&\vdots \\b_m &= a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n\end{aligned}$$

can be written in matrix form as:

$$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

or $\mathbf{b} = A\mathbf{x}$.

Questions for Linear Systems

We want to know

Questions for Linear Systems

We want to know

- Given a left hand side vector $\mathbf{b}$, how can we find a solution $\mathbf{x}$ to $\mathbf{b} = A\mathbf{x}$?

Questions for Linear Systems

We want to know

- Given a left hand side vector $\mathbf{b}$, how can we find a solution $\mathbf{x}$ to $\mathbf{b} = A\mathbf{x}$?
- Given a coefficient matrix A , what can we say about the number of solutions to $\mathbf{b} = A\mathbf{x}$, for any $\mathbf{b}$? How can we tell whether a system has one unique solution?

Elementary Row Operations

There are three types of elementary row operations that one can perform on matrix A and the vector $\mathbf{b}$ without changing the solutions set to $\mathbf{b} = A\mathbf{x}$:

Elementary Row Operations

There are three types of elementary row operations that one can perform on matrix A and the vector $\mathbf{b}$ without changing the solutions set to $\mathbf{b} = A\mathbf{x}$:

- Interchanging rows

Elementary Row Operations

There are three types of elementary row operations that one can perform on matrix A and the vector $\mathbf{b}$ without changing the solutions set to $\mathbf{b} = A\mathbf{x}$:

- Interchanging rows
- Multiplying each element of a row by the same non-zero scalar

Elementary Row Operations

There are three types of elementary row operations that one can perform on matrix A and the vector $\mathbf{b}$ without changing the solutions set to $\mathbf{b} = A\mathbf{x}$:

- Interchanging rows
- Multiplying each element of a row by the same non-zero scalar
- Change a row by adding it to a multiple of another row

Elementary Row Operations

There are three types of elementary row operations that one can perform on matrix A and the vector $\mathbf{b}$ without changing the solutions set to $\mathbf{b} = A\mathbf{x}$:

- Interchanging rows
- Multiplying each element of a row by the same non-zero scalar
- Change a row by adding it to a multiple of another row
- Note the similarities of these operations to the operations we learn to use when solving systems of equations without matrix algebra.

Using Elementary Row Operations to Solve a System of Equations: Row Echelon Form

A matrix is in *row echelon form* if the following three conditions are met

Using Elementary Row Operations to Solve a System of Equations: Row Echelon Form

A matrix is in *row echelon form* if the following three conditions are met

- All non-zero rows are above any row with all zeros.

Using Elementary Row Operations to Solve a System of Equations: Row Echelon Form

A matrix is in *row echelon form* if the following three conditions are met

- All non-zero rows are above any row with all zeros.
- The leading coefficient (the first non-zero element from the left) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.

Using Elementary Row Operations to Solve a System of Equations: Row Echelon Form

A matrix is in *row echelon form* if the following three conditions are met

- All non-zero rows are above any row with all zeros.
- The leading coefficient (the first non-zero element from the left) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.
- All entries in a column below a leading entry are zeros (this condition is implied by the other two).

Row Echelon Form: Examples

Which of the following matrices are in Row Echelon Form?

$$A = \begin{pmatrix} 4 & 2 \\ 0 & 7 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 10 \\ 0 & 8 & -3 \\ 0 & 0 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 8 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

Row Echelon Form: Examples

Which of the following matrices are in Row Echelon Form?

$$A = \begin{pmatrix} 4 & 2 \\ 0 & 7 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 10 \\ 0 & 8 & -3 \\ 0 & 0 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 8 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

Row Echelon Form: Examples

Which of the following matrices are in Row Echelon Form?

$$A = \begin{pmatrix} 4 & 2 \\ 0 & 7 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 & 10 \\ 0 & 8 & -3 \\ 0 & 0 & 6 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 8 \end{pmatrix}, D = \begin{pmatrix} 0 & 1 \\ 2 & 4 \end{pmatrix}$$

Note that D would be in row echelon form if we switched rows.

Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix} \quad A_1 = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

- Subtract the third row from the first row

Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

$$A_1 = \begin{pmatrix} 0 & 1 & 2 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix} \quad A_2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

- Subtract the third row from the first row
- Subtract 4 times the third row from the second row

Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

$$A_2 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 4 & 2 \\ 1 & 1 & 1 \end{pmatrix} \quad A_3 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & -6 \\ 1 & 1 & 1 \end{pmatrix}$$

- Subtract the third row from the first row
- Subtract 4 times the third row from the second row
- Subtract 4 times the first row from the second row

Row Echelon Form with Elementary Matrix Operations

To get a matrix in row echelon form we can perform elementary matrix operations on the rows. For example,

$$A_3 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & -6 \\ 1 & 1 & 1 \end{pmatrix} \quad A_4 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -6 \end{pmatrix}$$

- Subtract the third row from the first row
- Subtract 4 times the third row from the second row
- Subtract 4 times the first row from the second row
- Rearrange the rows to get row echelon form

Solving Linear System of Equations

We can solve a system of equations by writing the matrix

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{nn} & b_m \end{array} \right)$$

This is called the *augmented matrix*.

Solving Linear System of Equations: Example

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

and say the vector $\mathbf{b} = (1, 1, 1)'$. The the augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right)$$

Solving Linear System of Equations: Example continued

Performing the same matrix operations as before, we have

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 4 & 8 & 6 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right) \\ & \Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 2 & 1 \\ 0 & 4 & 2 & -3 \\ 1 & 1 & 1 & 1 \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \\ 1 & 1 & 1 & 1 \end{array} \right) \\ & \Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \end{array} \right) \end{aligned}$$

Solving Linear System of Equations: Example

Continue until LHS is the identity matrix

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -6 & -3 \end{array} \right) &\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \\ &\Rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & \frac{1}{2} \end{array} \right) \\ \left(\begin{array}{c} \frac{3}{2} \\ -1 \\ \frac{1}{2} \end{array} \right) &= \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right) \cdot \left(\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right) \Rightarrow \mathbf{x} = \left(\frac{3}{2}, -1, \frac{1}{2} \right) \end{aligned}$$

Reduced Row Echelon Form

We define a matrix to be in *reduced row echelon form* if it meets 3 conditions

Reduced Row Echelon Form

We define a matrix to be in *reduced row echelon form* if it meets 3 conditions

- Leading entries are all 1 and are the only non-zero entries in their column

Reduced Row Echelon Form

We define a matrix to be in *reduced row echelon form* if it meets 3 conditions

- Leading entries are all 1 and are the only non-zero entries in their column
- Leading entry of any lower row is to the right of the upper row

Reduced Row Echelon Form

We define a matrix to be in *reduced row echelon form* if it meets 3 conditions

- Leading entries are all 1 and are the only non-zero entries in their column
- Leading entry of any lower row is to the right of the upper row
- Any zero row is below all non-zero rows

Reduced Row Echelon Form

We define a matrix to be in *reduced row echelon form* if it meets 3 conditions

- Leading entries are all 1 and are the only non-zero entries in their column
- Leading entry of any lower row is to the right of the upper row
- Any zero row is below all non-zero rows

If we form the augmented matrix of A , reduce the left hand side to its reduced row echelon form through elementary row operations, then the remaining vector on the right hand side will be the solution to the system, although it may not be a unique solution.

Using the Determinant: Inverse and Cramer's Rule

Let A be a nonsingular matrix. Then,

Using the Determinant: Inverse and Cramer's Rule

Let A be a nonsingular matrix. Then,

- $A^{-1} = \frac{1}{\det A} \cdot \text{adj}A$

Using the Determinant: Inverse and Cramer's Rule

Let A be a nonsingular matrix. Then,

- $A^{-1} = \frac{1}{\det A} \cdot \text{adj}A$
- **Cramer's Rule** the unique solution $\mathbf{x} = (x_1, \dots, x_n)$ of the $n \times n$ system $A\mathbf{x} = \mathbf{b}$ is

$$x_i = \frac{\det B_i}{\det A}, \text{ for } i = 1, \dots, n,$$

where B_i is the matrix A with the right-hand side $\mathbf{b}$ replacing the i th column of A

Cramer's Rule: Example

Example: Consider the linear $IS - LM$ model, defined as

$$\begin{aligned}Y &= bY + (I^0 - ar) + G \\M_s &= mY + (M^0 - hr)\end{aligned}$$

where Y is the national product, b is marginal propensity to consume, a is the marginal efficiency of capital, $I = I^0 - ar$ is investment, m is money balances needed per dollar of transactions, G is government spending, M_s is the money supply and $M^0 - hr$ is speculative demand. Rearranging the terms yields

Cramer's Rule: Example

Example: Consider the linear $IS - LM$ model, defined as

$$\begin{aligned}Y &= bY + (I^0 - ar) + G \\M_s &= mY + (M^0 - hr)\end{aligned}$$

where Y is the national product, b is marginal propensity to consume, a is the marginal efficiency of capital, $I = I^0 - ar$ is investment, m is money balances needed per dollar of transactions, G is government spending, M_s is the money supply and $M^0 - hr$ is speculative demand. Rearranging the terms yields

$$\begin{aligned}sY + ar &= I^0 + G \\mY - hr &= M_s - M^0\end{aligned}$$

where $s = 1 - b$ is the marginal propensity to save.

Cramer's Rule: Example

We can rewrite the system as

$$\begin{pmatrix} I^0 + G \\ M_s - M^0 \end{pmatrix} = \begin{pmatrix} s & a \\ m & -h \end{pmatrix} \cdot \begin{pmatrix} Y \\ r \end{pmatrix}$$

By Cramer's rule, we have

$$Y = \frac{\begin{vmatrix} I^0 + G & a \\ M_s - M^0 & -h \end{vmatrix}}{\begin{vmatrix} s & a \\ m & -h \end{vmatrix}} = \frac{(I^0 + G)h + a(M_s - M^0)}{sh + am}$$
$$r = \frac{\begin{vmatrix} s & I^0 + G \\ m & M_s - M^0 \end{vmatrix}}{\begin{vmatrix} s & a \\ m & -h \end{vmatrix}} = \frac{(I^0 + G)m - s(M_s - M^0)}{sh + am}$$

Linear Independence

Linear Independence The vectors $v_1, \dots, v_m$ are *linearly dependent* if there exist scalars $q_1, \dots, q_m$, not all zero such that

$$\sum_{i=1}^m q_i v_i = 0$$

Linear Independence

Linear Independence The vectors $v_1, \dots, v_m$ are *linearly dependent* if there exist scalars $q_1, \dots, q_m$, not all zero such that

$$\sum_{i=1}^m q_i v_i = 0$$

The vectors $v_1, \dots, v_m$ are *linearly independent* if the only scalars $q_1, \dots, q_m$ such that

$$\sum_{i=1}^m q_i v_i = 0$$

are $q_1 = \dots = q_m = 0$.

Linear Independence

Linear Independence The vectors $v_1, \dots, v_m$ are *linearly dependent* if there exist scalars $q_1, \dots, q_m$, not all zero such that

$$\sum_{i=1}^m q_i v_i = 0$$

The vectors $v_1, \dots, v_m$ are *linearly independent* if the only scalars $q_1, \dots, q_m$ such that

$$\sum_{i=1}^m q_i v_i = 0$$

are $q_1 = \dots = q_m = 0$.

This is equivalent to the statement that none of the vectors can be written as a linear combination of the other vectors. Finally, note that we can use this definition for the columns or rows of a matrix.

Spanning Vectors

A set of vectors **spans** a vector space (a set of vectors obeying vector addition and scalar multiplication) if the vector space can be written as a linear combination of the vectors from the (spanning) set of vectors.

- The set of vectors $(1, 0)$ and $(0, 1)$ span 2-D space. Any vector on a Euclidean plane can be written as a linear combination of $(1, 0)$ and $(0, 1)$.

Spanning Vectors

A set of vectors **spans** a vector space (a set of vectors obeying vector addition and scalar multiplication) if the vector space can be written as a linear combination of the vectors from the (spanning) set of vectors.

- The set of vectors $(1, 0)$ and $(0, 1)$ span 2-D space. Any vector on a Euclidean plane can be written as a linear combination of $(1, 0)$ and $(0, 1)$.
- Spanning a vector space of $n - D$ requires n vectors that each have n elements.

Spanning Vectors

A set of vectors **spans** a vector space (a set of vectors obeying vector addition and scalar multiplication) if the vector space can be written as a linear combination of the vectors from the (spanning) set of vectors.

- The set of vectors $(1, 0)$ and $(0, 1)$ span 2-D space. Any vector on a Euclidean plane can be written as a linear combination of $(1, 0)$ and $(0, 1)$.
- Spanning a vector space of $n - D$ requires n vectors that each have n elements.
- Note that if one of these vectors can be written as a linear combination of the other vectors, then the vectors will not span $n - D$ space.

Spanning Vectors

A set of vectors **spans** a vector space (a set of vectors obeying vector addition and scalar multiplication) if the vector space can be written as a linear combination of the vectors from the (spanning) set of vectors.

- The set of vectors $(1, 0)$ and $(0, 1)$ span 2-D space. Any vector on a Euclidean plane can be written as a linear combination of $(1, 0)$ and $(0, 1)$.
- Spanning a vector space of $n - D$ requires n vectors that each have n elements.
- Note that if one of these vectors can be written as a linear combination of the other vectors, then the vectors will not span $n - D$ space.
- n linearly independent vectors must span the $n - D$ vector space.

Linear Independence: Example

Are these vectors linearly dependent

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$$

Linear Independence: Example

Are these vectors linearly dependent

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} = 0$$

Linear Independence: Example

Are these vectors linearly dependent

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} = 0$$

Are the following vectors linearly dependent?

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$$

Linear Independence: Example

Are these vectors linearly dependent

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} = 0$$

Are the following vectors linearly dependent?

$$3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$$

These vectors are linearly independent. The sum of these vectors only equals zero when $q_1 = q_2 = q_3 = 0$.

Rank

The rank of a matrix, $rk(A)$ is the number of linearly independent rows or columns in a matrix. (Note that the number of linearly independent rows is the same as the number of linearly independent columns). Therefore

Rank

The rank of a matrix, $rk(A)$ is the number of linearly independent rows or columns in a matrix. (Note that the number of linearly independent rows is the same as the number of linearly independent columns). Therefore

- $rk(A) \leq \min\{\text{number of rows of } A, \text{ number of columns of } A\}$

Rank

The rank of a matrix, $rk(A)$ is the number of linearly independent rows or columns in a matrix. (Note that the number of linearly independent rows is the same as the number of linearly independent columns). Therefore

- $rk(A) \leq \min\{\text{number of rows of } A, \text{number of columns of } A\}$
- A matrix has full rank if:

Rank

The rank of a matrix, $rk(A)$ is the number of linearly independent rows or columns in a matrix. (Note that the number of linearly independent rows is the same as the number of linearly independent columns). Therefore

- $rk(A) \leq \min\{\text{number of rows of } A, \text{ number of columns of } A\}$
- A matrix has full rank if:
 - $rk(A) = \min\{\text{number of rows of } A, \text{ number of columns of } A\}$

Rank and Row Echelon Form

For linear independence, we must have that all rows in the row echelon form are non-zero. If not, then the matrix will have linearly dependent rows. For example,

Rank and Row Echelon Form

For linear independence, we must have that all rows in the row echelon form are non-zero. If not, then the matrix will have linearly dependent rows. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

is linearly independent because its row echelon form is

$$A_2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -6 \end{pmatrix}$$

Rank and Row Echelon Form

For linear independence, we must have that all rows in the row echelon form are non-zero. If not, then the matrix will have linearly dependent rows. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

is linearly independent because its row echelon form is

$$A_2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -6 \end{pmatrix}$$

Note that if there is a row of all zeros in reduced row echelon form we know either that the row was originally all zeros or it was a linear combination of the other rows.

Characterizing the Number of Solutions to a Linear System

The locus of all points in (x_1, x_2) , which satisfy the linear equation $a_{11}x_1 + a_{12}x_2 = b_1$ is a straight line in the plane. Therefore, the solution (x_1, x_2) of the two linear equations in two unknowns

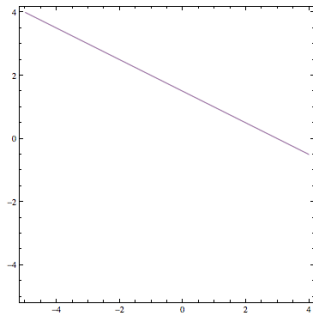
$$a_{11}x_1 + a_{12}x_2 = b_1$$

$$a_{21}x_1 + a_{22}x_2 = b_2$$

is a point that lies on both lines of these equations. Solving these systems is equivalent to finding where the lines given by the equations cross. If the lines are parallel, then there is no solution to the system. If the lines lie on top of each other, there are infinitely many solutions.

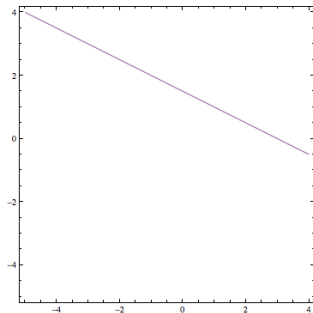
Solutions to a Linear System: Example 1

$$\begin{aligned}x_1 + 2x_2 &= 3 \\ 2x_1 + 4x_2 &= 6\end{aligned}$$



Solutions to a Linear System: Example 1

$$\begin{aligned}x_1 + 2x_2 &= 3 \\ 2x_1 + 4x_2 &= 6\end{aligned}$$

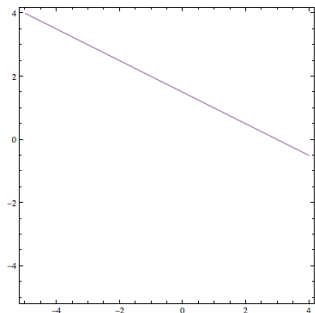


What is particular about this system of equations?

Solutions to a Linear System: Example 1

$$x_1 + 2x_2 = 3$$

$$2x_1 + 4x_2 = 6$$

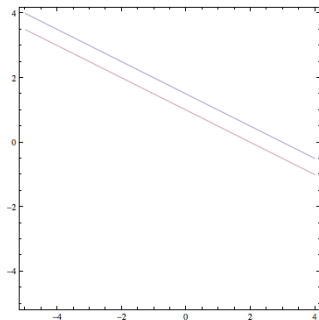


What is particular about this system of equations? If expressed in matrix terms the matrix $(A|\mathbf{b})$ has a rank of only 1, its rows are linearly dependent.

Characterizing the Number of Solutions to a Linear System: Example 2

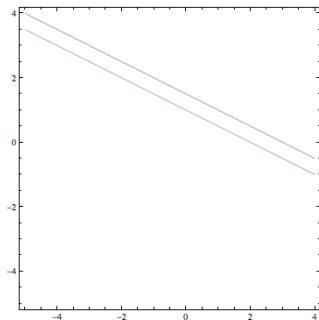
$$x_1 + 2x_2 = 3$$

$$2x_1 + 4x_2 = 4$$



Characterizing the Number of Solutions to a Linear System: Example 2

$$\begin{aligned}x_1 + 2x_2 &= 3 \\ 2x_1 + 4x_2 &= 4\end{aligned}$$

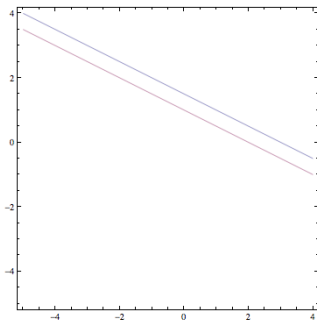


What is particular about this system of equations?

Characterizing the Number of Solutions to a Linear System: Example 2

$$x_1 + 2x_2 = 3$$

$$2x_1 + 4x_2 = 4$$

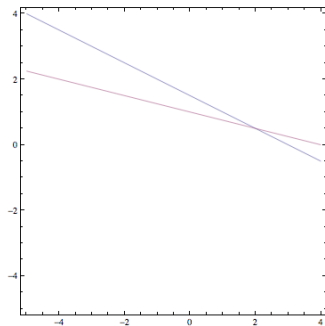


What is particular about this system of equations? If expressed in matrix terms, the matrix (A) has a rank of only 1, its rows are linearly dependent, which means the lines will be parallel.

Solutions to a Linear System: Example 3

$$x_1 + 2x_2 = 3$$

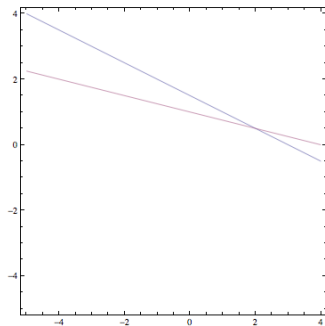
$$x_1 + 4x_2 = 4$$



Solutions to a Linear System: Example 3

$$x_1 + 2x_2 = 3$$

$$x_1 + 4x_2 = 4$$

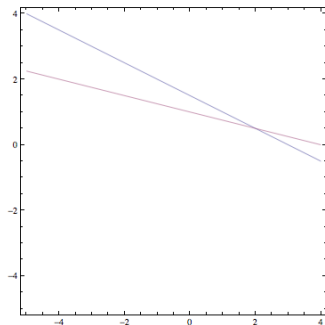


What is particular about this system of equations?

Solutions to a Linear System: Example 3

$$x_1 + 2x_2 = 3$$

$$x_1 + 4x_2 = 4$$



What is particular about this system of equations? If expressed in matrix terms, the matrix (A) has a rank of 2, its rows are linearly independent.

Solutions to a Linear System

Consider a system of m equations with n unknowns, characterized by $\mathbf{b} = A\mathbf{x}$. (This means that A will be an $m \times n$ matrix.) If $m < n$, then:

Solutions to a Linear System

Consider a system of m equations with n unknowns, characterized by $\mathbf{b} = A\mathbf{x}$. (This means that A will be an $m \times n$ matrix.) If $m < n$, then:

- For any given $\mathbf{b}$, $\mathbf{x} = A\mathbf{x}$ has zero or infinitely many solutions.

Solutions to a Linear System

Consider a system of m equations with n unknowns, characterized by $\mathbf{b} = A\mathbf{x}$. (This means that A will be an $m \times n$ matrix.) If $m < n$, then:

- For any given $\mathbf{b}$, $\mathbf{x} = A\mathbf{x}$ has zero or infinitely many solutions.
- If $rkA = m = rk(A|\mathbf{b})$, $\mathbf{b} = A\mathbf{x}$ has infinitely many solutions for every $\mathbf{b}$.

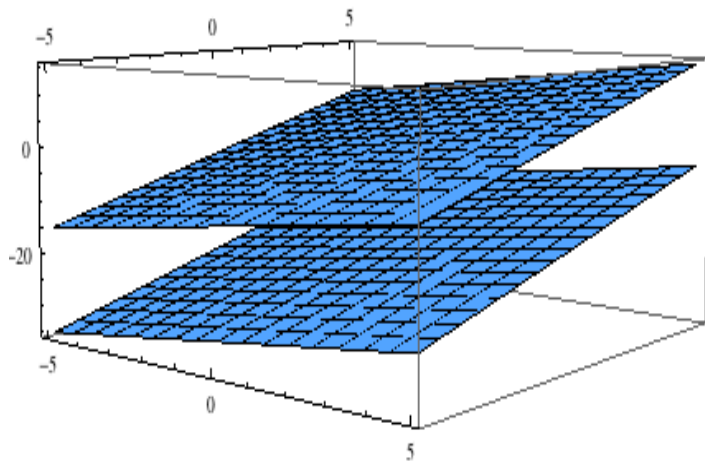
Solutions to a Linear System

Consider a system of m equations with n unknowns, characterized by $\mathbf{b} = A\mathbf{x}$. (This means that A will be an $m \times n$ matrix.) If $m < n$, then:

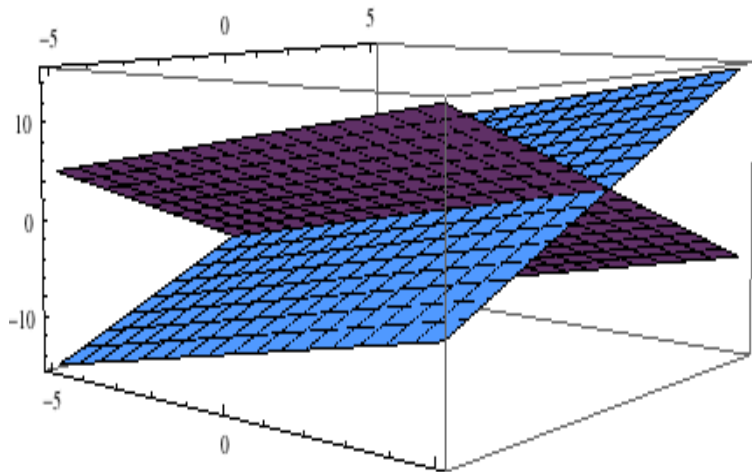
- For any given $\mathbf{b}$, $\mathbf{x} = A\mathbf{x}$ has zero or infinitely many solutions.
- If $rkA = m = rk(A|\mathbf{b})$, $\mathbf{b} = A\mathbf{x}$ has infinitely many solutions for every $\mathbf{b}$.

When a system has more unknowns than equations, it can never have a unique solution. For example, two planes in $\mathbb{R}^3$ cannot intersect at only one point. They either are parallel and have no solution or cross and have infinitely many solutions. If A has maximal rank of m then we know that the system has infinitely many solutions. For example in $\mathbb{R}^3$ the planes would cross yielding infinitely many solutions. We also know that the planes aren't identical but there are still an infinite amount of solutions.

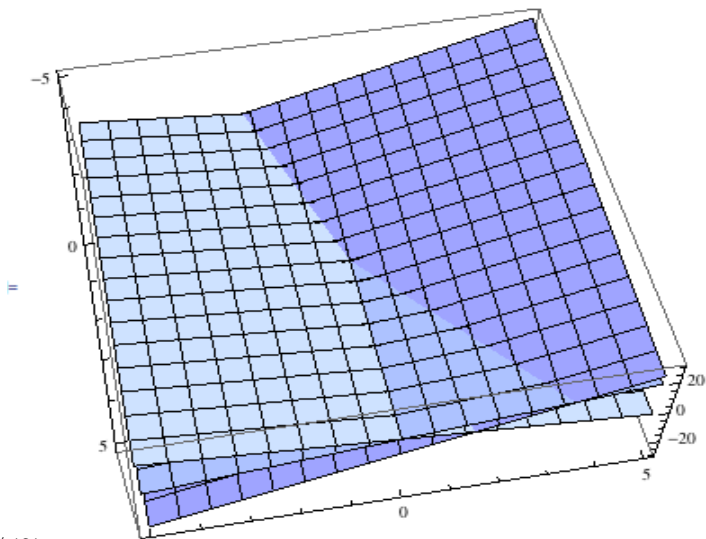
Planes in $\mathbb{R}^3$



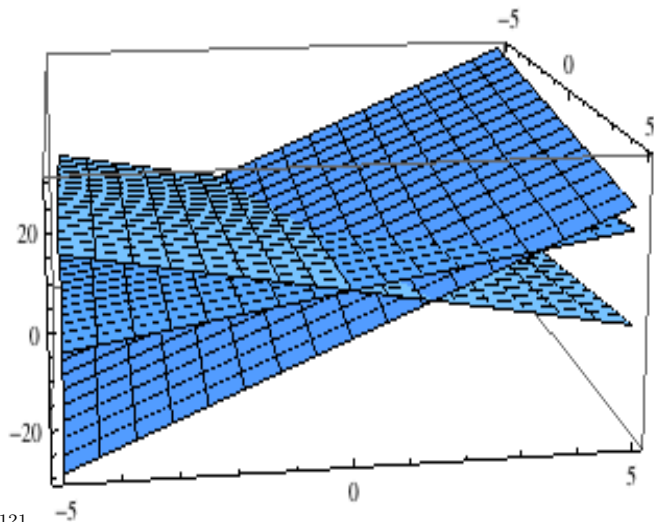
Planes in $\mathbb{R}^3$



Planes in $\mathbb{R}^3$



Planes in $\mathbb{R}^3$



Solutions to a Linear System

Now consider the case when $m > n$ (the number of equations exceeds the number of unknowns). Then

Solutions to a Linear System

Now consider the case when $m > n$ (the number of equations exceeds the number of unknowns). Then

- For any given $\mathbf{b}$, $\mathbf{b} = A\mathbf{x}$ has zero, one or infinitely many solutions

Solutions to a Linear System

Now consider the case when $m > n$ (the number of equations exceeds the number of unknowns). Then

- For any given $\mathbf{b}$, $\mathbf{b} = A\mathbf{x}$ has zero, one or infinitely many solutions
- If $rkA = n$, $\mathbf{b} = A\mathbf{x}$ has zero or one solution for every $\mathbf{b}$.

Solutions to a Linear System

Now consider the case when $m > n$ (the number of equations exceeds the number of unknowns). Then

- For any given $\mathbf{b}$, $\mathbf{b} = A\mathbf{x}$ has zero, one or infinitely many solutions
- If $rkA = n$, $\mathbf{b} = A\mathbf{x}$ has zero or one solution for every $\mathbf{b}$.

A system with more equations than unknowns may have zero, one or infinitely many solutions. For example, three lines in $\mathbb{R}^2$ can all not intersect (no solution), can intersect at one point (one solution) or can all be identical (infinitely many solutions). If we know that A has maximal rank of n then the system cannot have infinitely many solutions. We know this because all of the lines cannot be parallel or identical since they aren't linear complements. Still the system might have zero solutions if the lines do not cross at the same point.

Solutions to Linear System

Finally consider $m = n$, then

- For any given $\mathbf{b}$, $\mathbf{b} = A\mathbf{x}$ has zero, one or infinitely many solutions.
- If $\text{rk}A = m = n$, $\mathbf{b} = A\mathbf{x}$ has exactly one solution for every $\mathbf{b}$

A system with an equal number of equations and unknowns can have zero, one, or infinitely many solutions. For example, two lines in $\mathbb{R}^2$ can be parallel, intersect at one point or be identical. (Same for three planes in $\mathbb{R}^3$.) If we know that A has maximal rank then we know the system has a unique solution. In our example, we know that the lines are not parallel and must cross at a particular point.

Relationship between the Determinant and Linear System

An $n \times n$ square matrix A has full rank n if and only if $\det A \neq 0$. This means that the $n \times n$ system $\mathbf{b} = A\mathbf{x}$ has exactly one solution for each $\mathbf{b}$ if and only if $\det A \neq 0$. This means that if a matrix A is nonsingular it is equivalent to saying that $rkA = n$, which is equivalent to saying $\mathbf{b} = A\mathbf{x}$ has exactly one solution for each $\mathbf{b}$. Finally note that when $\det A = 0$ Cramer's rule fails.

Inverse Matrices and Linear Systems

Suppose we are given a $n \times n$ matrix A . Suppose that the inverse A^{-1} exists. Then we can multiply both sides of $\mathbf{b} = A\mathbf{x}$ by A^{-1} :

$$A^{-1}\mathbf{b} = A^{-1}A\mathbf{x} = I\mathbf{x} = \mathbf{x}$$

Thus $\mathbf{x} = A^{-1}\mathbf{b}$ has to be the unique solution to $\mathbf{b} = A\mathbf{x}$.

Conversely, it can also be shown that if there is a unique solution to $\mathbf{b} = A\mathbf{x}$, then the inverse A^{-1} has to exist. (For the proof see S and B page 167). Therefore, a square matrix is invertible if and only if it is nonsingular, etc...

The following statements are all equivalent for a square matrix A :

- A is non-singular
- All the columns and rows in A are linearly independent.
- A has full rank
- Exactly one solution $\mathbf{X}^*$ exists for each vector $\mathbf{Y}^*$
- A is invertible
- $\det(A) \neq 0$
- The row-echelon form of the matrix is upper triangular.
- The reduced row echelon form is the identity matrix

Calculating the Inverse Matrix by Elementary Row Operation

To calculate the inverse matrix, form the augmented matrix:

$$\left(\begin{array}{cccc|cccc} a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & 0 & 0 & \dots & 1 \end{array} \right)$$

Reduce the left hand side to the reduced row echelon form, and what remains on the right hand side will be the inverse matrix of A . In other words, using elementary row operations you can transform the matrix $(A|I)$ to $(I|A^{-1})$.

Quadratic Form

Consider the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, where $F = a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2$. We call this a *quadratic form* in $\mathbb{R}^2$. Notice that this can be expressed in matrix form as

$$F(\mathbf{x}) = (x_1 \ x_2) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} \\ \frac{1}{2}a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \mathbf{x}' A \mathbf{x}$$

where $x = (x_1, x_2)$ and A is unique and symmetric.

Quadratic Form in $\mathbb{R}^n$

The quadratic form in $\mathbb{R}^n$ is

$$F(\mathbf{x}) = \sum_{i,j=1}^n a_{ij}x_i x_j$$

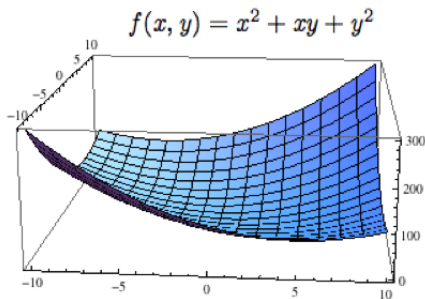
This can also be expressed in matrix form

$$F(\mathbf{x}) = (x_1 x_2 \dots x_n) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} & \dots & \frac{1}{2}a_{1n} \\ \frac{1}{2}a_{12} & a_{22} & \dots & \frac{1}{2}a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{2}a_{1n} & \frac{1}{2}a_{2n} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \mathbf{x}' A \mathbf{x}$$

Where $\mathbf{x} = (x_1, \dots, x_n)$ and A is unique and symmetric.

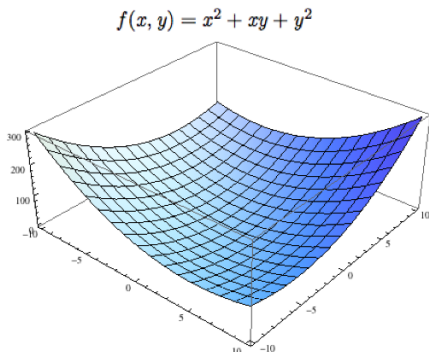
Definiteness

A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



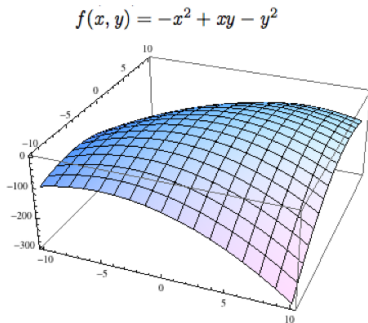
Definiteness

A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



Definiteness

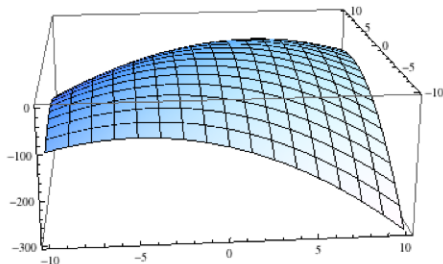
A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



Definiteness

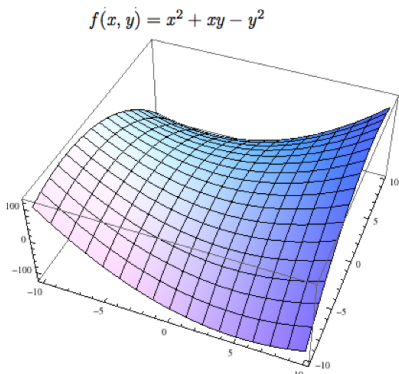
A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.

$$f(x, y) = -x^2 + xy - y^2$$



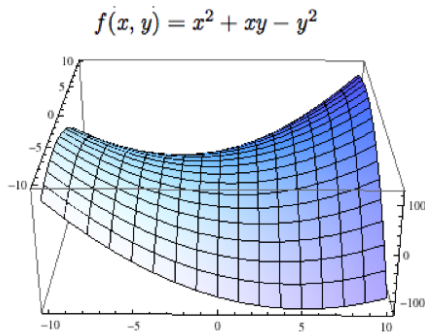
Definiteness

A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



Definiteness

A quadratic form has a critical point at $\mathbf{x}$, where it takes on the value 0. We can thus, classify quadratic forms by whether $\mathbf{x} = 0$ is a maximum, minimum, or neither. The properties of definiteness provide this information. To first get some intuition consider the following simple quadratics.



Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

- *Positive definite* if $\mathbf{x}'A\mathbf{x} > 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global minimum of the quadratic form given by A .

Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

- *Positive definite* if $\mathbf{x}'A\mathbf{x} > 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global minimum of the quadratic form given by A .
- *Positive Semidefinite* if $\mathbf{x}'A\mathbf{x} \geq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global minimum, but not a unique global one, of the quadratic form given by A .

Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

- *Positive definite* if $\mathbf{x}'A\mathbf{x} > 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global minimum of the quadratic form given by A .
- *Positive Semidefinite* if $\mathbf{x}'A\mathbf{x} \geq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global minimum, but not a unique global one, of the quadratic form given by A .
- *Negative Definite* if $\mathbf{x}'A\mathbf{x} < 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global maximum of the quadratic form given by A .
- *Negative Semidefinite* if $\mathbf{x}'A\mathbf{x} \leq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global maximum, but not a unique global one, of the quadratic form given by A .

Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

- *Positive definite* if $\mathbf{x}'A\mathbf{x} > 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global minimum of the quadratic form given by A .
- *Positive Semidefinite* if $\mathbf{x}'A\mathbf{x} \geq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global minimum, but not a unique global one, of the quadratic form given by A .
- *Negative Definite* if $\mathbf{x}'A\mathbf{x} < 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global maximum of the quadratic form given by A .
- *Negative Semidefinite* if $\mathbf{x}'A\mathbf{x} \leq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global maximum, but not a unique global one, of the quadratic form given by A .

Definiteness

Let A be an $n \times n$ symmetric matrix. Then A is:

- *Positive definite* if $\mathbf{x}'A\mathbf{x} > 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global minimum of the quadratic form given by A .
- *Positive Semidefinite* if $\mathbf{x}'A\mathbf{x} \geq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global minimum, but not a unique global one, of the quadratic form given by A .
- *Negative Definite* if $\mathbf{x}'A\mathbf{x} < 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a unique global maximum of the quadratic form given by A .
- *Negative Semidefinite* if $\mathbf{x}'A\mathbf{x} \leq 0 \forall \mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is a global maximum, but not a unique global one, of the quadratic form given by A .
- *Indefinite* if $\mathbf{x}'A\mathbf{x} > 0$ for some $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$ and $\mathbf{x}'A\mathbf{x} < 0$ for some other $\mathbf{x} \neq \mathbf{0}$ in $\mathbb{R}^n$. That is $\mathbf{x} = \mathbf{0}$ is neither a maximum or minimum, of the quadratic form given by A .

Principal Minors

Let A be an $n \times n$ matrix. A $k \times k$ submatrix of A obtained by deleting any $n - k$ columns and the same $n - k$ rows from A is called a k th-order *principal submatrix* of A . The determinant of the $k \times k$ principal submatrix is called the k th order *principal minor* of A .

Principal Minors: Example

List all the principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Principal Minors: Example

List all the principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- There is 1 third order principal minor of A , $\det(A)$.

Principal Minors: Example

List all the principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

■ There is 1 third order principal minor of A , $\det(A)$.

■ There are three second order principal minors: $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$

$$\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

Principal Minors: Example

List all the principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- There is 1 third order principal minor of A , $\det(A)$.
- There are three second order principal minors: $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$,
 $\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$, $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$
- There are also three first order principal minors a_{11} , a_{22} , and a_{33} .

Leading Principal Minors

The *leading principal minor* is the determinant of the *leading principal submatrix* obtained by deleting the last $n - k$ rows and columns of an $n \times n$ matrix A . For example list the first, second, and third order leading principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Leading Principal Minors

The *leading principal minor* is the determinant of the *leading principal submatrix* obtained by deleting the last $n - k$ rows and columns of an $n \times n$ matrix A . For example list the first, second, and third order leading principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- The first order leading principal minor is $\begin{vmatrix} a_{11} \end{vmatrix}$

Leading Principal Minors

The *leading principal minor* is the determinant of the *leading principal submatrix* obtained by deleting the last $n - k$ rows and columns of an $n \times n$ matrix A . For example list the first, second, and third order leading principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- The first order leading principal minor is $\begin{vmatrix} a_{11} \end{vmatrix}$
- The second order leading principal minor is: $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$

Leading Principal Minors

The *leading principal minor* is the determinant of the *leading principal submatrix* obtained by deleting the last $n - k$ rows and columns of an $n \times n$ matrix A . For example list the first, second, and third order leading principal minors of the 3×3 matrix:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- The first order leading principal minor is $\begin{vmatrix} a_{11} \end{vmatrix}$
- The second order leading principal minor is: $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$
- There is one third order leading principal minor of A , $\det(A)$.

Testing for Definiteness

We can test for the definiteness of the matrix in the following fashion:

Testing for Definiteness

We can test for the definiteness of the matrix in the following fashion:

- A is positive definite iff all of its n leading principal minors are strictly positive.

Testing for Definiteness

We can test for the definiteness of the matrix in the following fashion:

- A is positive definite iff all of its n leading principal minors are strictly positive.
- A is negative definite iff all of its n leading principal minors alternate in sign, where $|A_1| < 0$, $|A_2| > 0$, $|A_3| < 0$, etc...

Testing for Definiteness

We can test for the definiteness of the matrix in the following fashion:

- A is positive definite iff all of its n leading principal minors are strictly positive.
- A is negative definite iff all of its n leading principal minors alternate in sign, where $|A_1| < 0$, $|A_2| > 0$, $|A_3| < 0$, etc...
- If some k th order leading principal minor of A is nonzero but does not fit either of the above sign pattern, then A is indefinite.

Testing for Semidefiniteness.

In the previous slide, verifying if A is positive definite or negative definite depends on strict inequalities. If these inequalities are relaxed we know that the A is not positive or negative definite. It may still be positive or negative semidefinite. To verify this, we look for the following conditions

Testing for Semidefiniteness.

In the previous slide, verifying if A is positive definite or negative definite depends on strict inequalities. If these inequalities are relaxed we know that the A is not positive or negative definite. It may still be positive or negative semidefinite. To verify this, we look for the following conditions

- A is positive semidefinite iff *every* principal minor of A is ≥ 0

Testing for Semidefiniteness.

In the previous slide, verifying if A is positive definite or negative definite depends on strict inequalities. If these inequalities are relaxed we know that the A is not positive or negative definite. It may still be positive or negative semidefinite. To verify this, we look for the following conditions

- A is positive semidefinite iff *every* principal minor of A is ≥ 0
- A is negative semidefinite iff *every* principal minor of A of odd order is ≤ 0 and every principal minor of even order is ≥ 0 .

Testing for Semidefiniteness.

In the previous slide, verifying if A is positive definite or negative definite depends on strict inequalities. If these inequalities are relaxed we know that the A is not positive or negative definite. It may still be positive or negative semidefinite. To verify this, we look for the following conditions

- A is positive semidefinite iff *every* principal minor of A is ≥ 0
- A is negative semidefinite iff *every* principal minor of A of odd order is ≤ 0 and every principal minor of even order is ≥ 0 .

Notice that for determining semi-definiteness, we can no longer check just the leading principal minors, but we must check all principal minors.

Eigenvalues

Let A be a square matrix. An *eigenvalue* is a number λ such that

$$(A - \lambda I)\mathbf{x} = 0, \text{ for some } \mathbf{x} \neq 0$$

Also notice that this implies that an eigenvalue is a number λ such that $A\mathbf{x} = \lambda\mathbf{x}$.

Eigenvalues and Singularity

For the system $(A - \lambda I)\mathbf{x} = 0$, note that $\mathbf{x} = 0$ is always a solution. The definition of an eigenvalue requires that there is another solution to $(A - \lambda I)\mathbf{x} = 0$. This means that the system $(A - \lambda I)\mathbf{x} = 0$ has more than one solution. What does this tell us about the matrix $(A - \lambda I)$?

- We know that when the eigenvalue λ is subtracted from the diagonal elements of A , the resulting matrix is singular. Therefore, we know that $\det(A - \lambda I) = 0$

Eigenvalues and Singularity

For the system $(A - \lambda I)\mathbf{x} = 0$, note that $\mathbf{x} = 0$ is always a solution. The definition of an eigenvalue requires that there is another solution to $(A - \lambda I)\mathbf{x} = 0$. This means that the system $(A - \lambda I)\mathbf{x} = 0$ has more than one solution. What does this tell us about the matrix $(A - \lambda I)$?

- We know that when the eigenvalue λ is subtracted from the diagonal elements of A , the resulting matrix is singular. Therefore, we know that $\det(A - \lambda I) = 0$
- Also we know that A is non-singular if and only if 0 is not an eigenvalue of A .

Eigenvalue: Example

Find the eigenvalues of the matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix}$$

To find the eigenvalues we must solve for λ in the following equation

$$\left| \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

Eigenvalue: Example Continued

$$\left| \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{pmatrix} \right| = 0$$

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0$$

$$-(\lambda - 4)(\lambda^2 - 4\lambda + 1) = 0$$

So one eigenvalue is $\lambda = 4$.

Eigenvalue: Example Continued

$$\left| \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{pmatrix} \right| = 0$$

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0$$

$$-(\lambda - 4)(\lambda^2 - 4\lambda + 1) = 0$$

So one eigenvalue is $\lambda = 4$. The other is obtained by using the quadratic formula

$$\lambda = 4 \pm \frac{\sqrt{(-4)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1} = 2 \pm \sqrt{3}$$

The eigenvalues are $\lambda = \{2 - \sqrt{3}, 2 + \sqrt{3}, 4\}$.

Eigenvectors

Given an eigenvalue λ , an *eigenvector* associated with λ is a non-zero vector $\mathbf{x}$ such that $(A - \lambda I)\mathbf{x} = 0$. Notice that eigenvectors are only defined up to a scalar: If $(A - \lambda I)\mathbf{x} = 0$, then also $(A - \lambda I)2\mathbf{x} = 0$. So all multiples of $\mathbf{x}$ are also eigenvectors associated with λ .

Intuition of Eigenvector

Eigenvectors are vectors that, when multiplied by A , end up as scalar multiples of themselves. The transformed vector points in the same direction as the original vector, but may vary in length. Consider the matrix.

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\text{Then } A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix}$$

- $A \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector for the eigenvalue of 4, since

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}.$$

Intuition Continued

Something like $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is not since $A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

Eigenvectors: Example

Find the eigenvectors of the matrix

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

What is the first step?

Eigenvectors: Example

Find the eigenvectors of the matrix

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

What is the first step? First we must find the eigenvalues of the matrix. The determinant of the matrix $(A - \lambda I)$ is

$$-\lambda(2-\lambda)(3-\lambda)+2(2-\lambda) = -\lambda(6-5\lambda+\lambda^2)+4+2\lambda = -8\lambda+5\lambda^2-\lambda^3+4$$

Thus we must find λ such that the determinant is equal to zero or

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)(\lambda - 2) = 0$$

This means that the eigenvalues are $\{1, 2, 2\}$.

Eigenvectors: Example

We have calculated the eigenvalues to be $\{1, 2, 2\}$. Note that each eigenvector corresponds to an eigenvalue. First consider the eigenvalue $\lambda = 1$. The matrix $A - \lambda I$ is then

$$\begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$$

This implies that the system of equations corresponding to $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is

$$x_1 = -2x_3$$

$$x_1 = -x_2 - x_3$$

$$x_1 = -2x_3$$

Eigenvectors: Example

Using the system of equations we can solve for the eigenvector.

$$x_1 = -2x_3$$

$$x_1 = -x_2 - x_3$$

$$x_1 = -2x_3$$

This implies that $-2x_3 = x_1 = -x_2 - x_3 \implies x_3 = x_2$ and we have

And so

$$x_1 = -2x_3$$

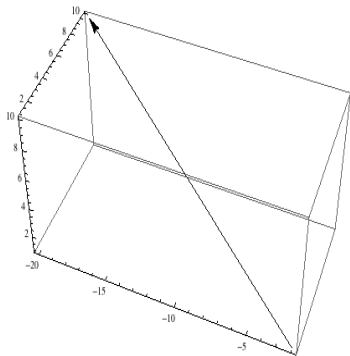
$$x_2 = x_3$$

$$x_3 = x_3$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} x_3$$

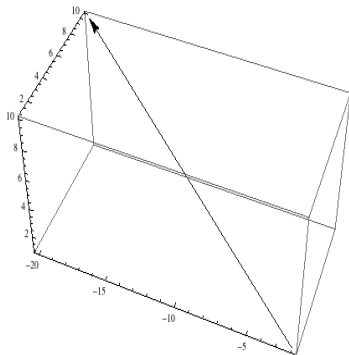
Eigenvector: Example

The eigenvector is defined up to a scalar. So we can let x_3 be any value we want, say $x_3 = s$ and trace out the line in 3 dimensional space. The direction of this line is the eigenvector of $\lambda = 1$. For each point on this line it is the case $(A - \lambda I)\mathbf{x} = 0$.



Eigenvector: Example

The eigenvector is defined up to a scalar. So we can let x_3 be any value we want, say $x_3 = s$ and trace out the line in 3 dimensional space. The direction of this line is the eigenvector of $\lambda = 1$. For each point on this line it is the case $(A - \lambda I)\mathbf{x} = 0$. Are we done?



Eigenvector: Example

We know one eigenvector, now we need to find the eigenvector that corresponds to $\lambda = 2$. First, we find

Eigenvector: Example

We know one eigenvector, now we need to find the eigenvector that corresponds to $\lambda = 2$. First, we find

$$(A - \lambda I) = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Eigenvector: Example

We know one eigenvector, now we need to find the eigenvector that corresponds to $\lambda = 2$. First, we find

$$(A - \lambda I) = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

This implies

$$x_1 = -x_3$$

$$x_1 = -x_3$$

$$x_1 = -x_3$$

Which implies

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ ? \\ -1 \end{pmatrix} x_1$$

Eigenvector: Example

Since the system that corresponds to $\lambda = 2$ is independent of x_2 , we can let both x_1 and x_2 be anything we want. Say $x_1 = s$ and $x_2 = t$. Then we can write the solution to the system as

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} s + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} t$$

The eigenvectors that correspond to $\lambda = 2$ are $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and

$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and their linear combinations.

Eigenvector: Example

To check that the eigenvectors and their linear combinations correspond to $\lambda = 2$ we can just verify that for each eigenvector $(A - \lambda I)\mathbf{x} = 0$.

$$(A - \lambda I)\mathbf{x} = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 + 0 + 2 \\ 1 + 0 - 1 \\ 1 + 0 - 1 \end{pmatrix} = 0$$

$$(A - \lambda I)\mathbf{x} = \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 + 0 + 0 \\ 0 + 0 + 0 \\ 0 + 0 + 0 \end{pmatrix} = 0$$

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the eigenvectors that correspond to A and the system $(A - \lambda I)\mathbf{x}$ for each λ

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the eigenvectors that correspond to A and the system $(A - \lambda I)\mathbf{x}$ for each λ

- Chose a λ and compute $(A - \lambda I)$

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the eigenvectors that correspond to A and the system $(A - \lambda I)\mathbf{x}$ for each λ

- Chose a λ and compute $(A - \lambda I)$
- Identify the system of equations that corresponds to $(A - \lambda I)\mathbf{x} = 0$

Steps for finding the Eigenvalue and Eigenvector

Steps for finding the eigenvalues that correspond to the $n \times n$ matrix A .

- Recognize that $(A - \lambda I)$ is singular and thus its determinate is equal to zero
- Calculate the determinate, which yields an n th order polynomial and solve for λ

Steps for finding the eigenvectors that correspond to A and the system $(A - \lambda I)\mathbf{x}$ for each λ

- Chose a λ and compute $(A - \lambda I)$
- Identify the system of equations that corresponds to $(A - \lambda I)\mathbf{x} = 0$
- Solve for all of the elements of the vector $\mathbf{x}$ so that they are defined in terms of one element of this vector. Use this to define the eigenvectors.

Number of Eigenvalues and Eigenvectors

Finally note that the equation defining λ is a for a $n \times n$ matrix is a n order polynomial equation. From our first example,

$$\left| \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{pmatrix} \right| = 0$$

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0$$

This also means that there are n roots and thus n eigenvalues for a $n \times n$ matrix, though some may not be real. Each eigenvalue has a corresponding eigenvector. If an eigenvalue is repeated k times then it will have k corresponding eigenvectors.

Leslie Population Model: Application of Eigenvalues and Eigenvectors

This model describes population dynamics of an organism that lives for two years. Let b_1 be the birthrate of individuals in their first year and let b_2 be the birthrate of individuals in their second year. Let d_1 be the death rate of individuals of first year individuals so that $(1 - d_1)$ of the first year individuals survive to year two. Let x_n and y_n denote the number of first and second year individuals in the year n . Given an initial distribution of first and second year individuals, we want to know how many individuals will exist in year n . This system is described by the two equations:

$$x_{n+1} = b_1 x_n + b_2 y_n$$

$$y_{n+1} = (1 - d_1) x_n$$

Leslie Population Model: Example

Consider the coupled system of difference equation generated by a Leslie model with $b_1 = 1$, $b_2 = 4$, and $d_1 = 0.5$

$$x_{n+1} = 1x_n + 4y_n$$

$$y_{n+1} = 0.5x_n$$

Leslie Population Model: Example

To uncouple system we need to change the coordinates of the system (See SB 586-588). In this case the coordinate change that will uncouple the system is:

$$\begin{aligned}X &= \frac{1}{6}x + \frac{1}{3}y \\Y &= -\frac{1}{6}x + \frac{2}{3}y\end{aligned}$$

The inverse transformation is

$$\begin{aligned}x &= 4X - 2Y \\(\text{so that } x &= 4(\frac{1}{6}x + \frac{1}{3}y) - 2(-\frac{1}{6}x + \frac{2}{3}y) = x) \\y &= X + Y\end{aligned}$$

The choice of this transformation arises from a consideration of the eigenvalues and eigenvectors of the matrix representation of the system defined above.

Leslie Population Model: Example

In matrix form the transitions are

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & \frac{1}{3} \\ -\frac{1}{6} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

Note that the two matrices are inverse of each other.

Leslie Population Model: Example

Now use these transformations to change the system from x, y variables to X, Y variables:

$$\begin{aligned}X_{n+1} &= \frac{1}{6}x_{n+1} + \frac{1}{3}y_{n+1} = \frac{1}{6}(x_n + 4y_n) + \frac{1}{3}\left(\frac{1}{2}(x_n)\right) \\Y_{n+1} &= -\frac{1}{6}x_{n+1} + \frac{2}{3}y_{n+1} = -\frac{1}{6}(x_n + 4y_n) + \frac{2}{3}\left(\frac{1}{2}(x_n)\right)\end{aligned}$$

Leslie Population Model: Example

Now use these transformations to change the system from x, y variables to X, Y variables:

$$\begin{aligned}X_{n+1} &= \frac{1}{6}x_{n+1} + \frac{1}{3}y_{n+1} = \frac{1}{6}(x_n + 4y_n) + \frac{1}{3}\left(\frac{1}{2}(x_n)\right) \\Y_{n+1} &= -\frac{1}{6}x_{n+1} + \frac{2}{3}y_{n+1} = -\frac{1}{6}(x_n + 4y_n) + \frac{2}{3}\left(\frac{1}{2}(x_n)\right)\end{aligned}$$

Simplifying and using the inverse change of coordinate system we get:

$$\begin{aligned}X_{n+1} &= \frac{1}{3}(x_n) + \frac{2}{3}(y_n) = \frac{1}{3}(4X_n - 2Y_n) + \frac{2}{3}(X_n + Y_n) = 2X_n \\Y_{n+1} &= -\frac{1}{6}(x_n) - \frac{2}{3}(y_n) = \frac{1}{6}(4X_n - 2Y_n) - \frac{2}{3}(X_n + Y_n) = -Y_n\end{aligned}$$

Leslie Population Model: Example

Now use these transformations to change the system from x, y variables to X, Y variables:

$$\begin{aligned}X_{n+1} &= \frac{1}{6}x_{n+1} + \frac{1}{3}y_{n+1} = \frac{1}{6}(x_n + 4y_n) + \frac{1}{3}\left(\frac{1}{2}(x_n)\right) \\Y_{n+1} &= -\frac{1}{6}x_{n+1} + \frac{2}{3}y_{n+1} = -\frac{1}{6}(x_n + 4y_n) + \frac{2}{3}\left(\frac{1}{2}(x_n)\right)\end{aligned}$$

Simplifying and using the inverse change of coordinate system we get:

$$\begin{aligned}X_{n+1} &= \frac{1}{3}(x_n) + \frac{2}{3}(y_n) = \frac{1}{3}(4X_n - 2Y_n) + \frac{2}{3}(X_n + Y_n) = 2X_n \\Y_{n+1} &= -\frac{1}{6}(x_n) - \frac{2}{3}(y_n) = \frac{1}{6}(4X_n - 2Y_n) - \frac{2}{3}(X_n + Y_n) = -Y_n\end{aligned}$$

The system is uncoupled and can be solved as two one-dimensional equations.

$$\begin{aligned}X_n &= 2^n c_1 \\(\text{e.g. } X_2 &= 2(X_1) \text{ and } X_1 = 2(X_0) \implies X_2 = 2(2(X_0))) \\Y_n &= (-1)^n c_2\end{aligned}$$

Leslie Population Model: Example

Finally we use the change in coordinates to express the system in terms of x_n and y_n .

$$\begin{aligned}x_n &= 4X_n - 2Y_n = 4 \cdot 2^n c_1 - 2 \cdot (-1)^n c_2 \\y_n &= X_n + Y_n = 2^n c_1 + (-1)^n c_2\end{aligned}$$

We can solve for the constants c_1 and c_2 , if we are given the initial endowments x_0 and y_0 . Since these are the endowments when $n = 0$, we can solve for the constants using the system of equations

$$\begin{aligned}x_0 &= 4c_1 - 2c_2 \\y_0 &= c_1 + c_2\end{aligned}$$

Abstract two-dimensional systems

Lets redo the entire process for an abstract system of difference equations

$$\mathbf{z}_{n+1} = A\mathbf{z}_n$$

Now we will follow the same steps using P and P^{-1} for the change of coordinate matrices. This will show the role that eigenvalues and eigenvectors play in changing the coordinates to uncouple the system. We write the change in coordinates matrices as

$$\mathbf{z} = P\mathbf{Z} \text{ or } \mathbf{Z} = p^{-1}\mathbf{z}$$

where the original variables are written as $\mathbf{z}$ and the transformed variables are written as $\mathbf{Z}$.

Abstract two-dimensional systems

Original difference and transformation equations:

$$\mathbf{z}_{n+1} = A\mathbf{z}_n, \mathbf{z} = P\mathbf{Z} \text{ and } \mathbf{Z} = P^{-1}\mathbf{z}$$

The steps for uncoupling this system are:

- $\mathbf{Z}_{n+1} = P^{-1}\mathbf{z}_{n+1}$ (Applying $\mathbf{Z} = P^{-1}\mathbf{z}$)
- $= P^{-1}(A\mathbf{z}_n)$ (Applying $\mathbf{z}_{n+1} = A\mathbf{z}_n$)
- $= (P^{-1}A)\mathbf{z}_n$ (Simplifying)
- $= (P^{-1}A)(P\mathbf{Z}_n)$ (applying $\mathbf{z} = P\mathbf{Z}$)
- $= (P^{-1}AP)\mathbf{Z}_n$ (simplifying)

Each step here corresponds to a step in the previous example.

This that if we change the coordinates using a linear transformation $\mathbf{z} = P\mathbf{Z}$, the coefficient matrix A is transformed to the matrix $P^{-1}AP$.

Abstract two-dimensional systems

We want to choose a transformation P such that the system $\mathbf{Z}_{n+1} = (P^{-1}AP)\mathbf{Z}_n$ is as simple as possible. If $P^{-1}AP$ were a diagonal matrix then the system would be uncoupled and easily solved. Focusing on the two-dimensional case, we can identify what kind of matrix P will lead to a diagonal $P^{-1}AP$. Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be the two columns of the 2×2 matrix P , thus $P = [\mathbf{p}_1 \mathbf{p}_2]$ and write the diagonal matrix D as $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$. Now $P^{-1}AP = D \implies AP = PD$ for an invertible matrix P . We can write this as

$$\begin{aligned} A[\mathbf{p}_1 \mathbf{p}_2] &= [\mathbf{p}_1 \mathbf{p}_2] \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \\ \implies [A\mathbf{p}_1 \ A\mathbf{p}_2] &= [\lambda_1 \mathbf{p}_1 \ \lambda_2 \mathbf{p}_2] \\ A\mathbf{p}_1 &= \lambda_1 \mathbf{p}_1 \text{ and } A\mathbf{p}_2 = \lambda_2 \mathbf{p}_2 \end{aligned}$$

Abstract two-dimensional systems

Note that $A\mathbf{p}_1 = \lambda_1\mathbf{p}_1$ and $A\mathbf{p}_2 = \lambda_2\mathbf{p}_2$ can be rewritten as $(A - \lambda_1 I)\mathbf{p}_1 = 0$ and $(A - \lambda_2 I)\mathbf{p}_2 = 0$. What does this tell us about λ_1 , λ_2 , $\mathbf{p}_1$, and $\mathbf{p}_2$?

Abstract two-dimensional systems

Note that $A\mathbf{p}_1 = \lambda_1\mathbf{p}_1$ and $A\mathbf{p}_2 = \lambda_2\mathbf{p}_2$ can be rewritten as $(A - \lambda_1 I)\mathbf{p}_1 = 0$ and $(A - \lambda_2 I)\mathbf{p}_2 = 0$. What does this tell us about λ_1 , λ_2 , $\mathbf{p}_1$, and $\mathbf{p}_2$?

λ_1 and λ_2 are the eigenvalues that correspond to the matrix A . $\mathbf{p}_1$, and $\mathbf{p}_2$ are the eigenvectors that correspond to the matrix A and the eigenvalues λ_1 and λ_2 . This result can be generalized for a k -dimensional system. (See SB 592.)

Diagonalizability

A square matrix A is called *diagonalizable* if there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix. We state without proof the following theorem (For proof see SB appendix of chap 23). Let A be a $n \times n$ matrix and assume that A has n linearly independent eigenvectors. Define the matrices

$$D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}, \quad P = \begin{pmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{pmatrix}$$

where $\mathbf{p}_i$ is the i th linearly independent eigenvector. Then $D = P^{-1}AP$ and $PDP^{-1} = A$.

Example of a Non-Diagonalizable Matrix

Consider the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$\det(A - \lambda I) = (1 - \lambda)^2$ Setting $\det(A - \lambda I) = 0$ and solving for λ we find that the repeated eigenvalue is $\lambda = 1$. To find the associated eigenvector, we solve

$$(A - 1I)\mathbf{x} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{x} = \mathbf{0}$$

We get $\mathbf{x} = \begin{pmatrix} x_1 \\ 0 \end{pmatrix}$, which can be written as $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} s$. In this case there is only one linearly independent eigenvector. Since we cannot find two linearly independent eigenvectors A is not diagonalizable.

Markov Chains

Assume k states of the world, which we label $1, 2, \dots, k$. Let the probability that we move to state j from state i be p_{ij} , and call it the *transition probability*. Then the *transition probability matrix* of the Markov chain is the matrix

$$P = \begin{pmatrix} p_{11} & \dots & p_{1k} \\ \vdots & \ddots & \vdots \\ p_{k1} & \dots & p_{kk} \end{pmatrix}$$

For this matrix the rows are the current state of the world and the columns are the states into which we can move. Also the elements in the rows must add up to 1 or the system would have a positive probability of moving into an undefined state.

Markov Chains: Example

Say that there are three states of the world: rainy, overcast, and sunny. Let state 1 be rainy, 2 be cloudy, and 3 be sunny. Consider the transition matrix.

$$P = \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix}$$

For example, if it is rainy the probability that it remains rainy is .8. Also, if it is sunny, the probability that it becomes cloudy is .6.

Markov Chains: Example of Two Period Transition Matrix

What if we want to know of a state in two periods?

Markov Chains: Example of Two Period Transition Matrix

What if we want to know of a state in two periods? We can create a new transition matrix P_2 , and $P_2 = PP$. Similarly it can be shown that $P_n = P^n$.

Markov Chains: Example of Two Period Transition Matrix

What if we want to know of a state in two periods? We can create a new transition matrix P_2 , and $P_2 = PP$. Similarly it can be shown that $P_n = P^n$. For example, we can ask what is the probability that it will be cloudy the day after tomorrow?

$$P = \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} = \begin{pmatrix} .69 & .16 & .15 \\ .40 & .37 & .23 \\ .38 & .26 & .36 \end{pmatrix}$$

Therefore, there is a 0.16 chance that if it is rainy today, then it will be cloudy the day after tomorrow.

Markov Chains: Probability Vector

Implicitly we have been assuming that we know with certainty the state of the world today. If we don't know the state of the world today we can define an initial vector of probabilities for today defined as $\mathbf{x}_0 = (p_1, p_2, \dots, p_n)$. Then the probabilities of the state of the world tomorrow can be defined as $\mathbf{x}_0 P = \mathbf{x}_1$.

Generally, a vector $\mathbf{x}$ with $0 \leq x_i \leq 1$ for all i and $\sum_{i=1}^n x_i = 1$ is called a *probability vector*. For example,

$$\begin{aligned}\mathbf{x}_2 = \mathbf{x}_1 P = \mathbf{x}_0 P^2 &= \begin{pmatrix} .3 & .6 & .1 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} \begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix} \\ &= \begin{pmatrix} .485 & .296 & .219 \end{pmatrix}\end{aligned}$$

There is a 0.296 chance that it will be cloudy day after tomorrow.

Steady State

A transition matrix P is called *regular* if P^n has only strictly positive entries for some integer n . A *steady state* vector $\mathbf{q}$ of a transition matrix P is a probability vector that satisfies the equation $\mathbf{q}P = \mathbf{q}$. Notice that this implies $P'\mathbf{q}' = \mathbf{q}'$. For an eigenvalue of $\lambda = 1$ this also implies $P'\mathbf{q}' = \lambda\mathbf{q}'$ or $(P' - \lambda I)\mathbf{q}' = \mathbf{0}$. If P is a regular transition matrix the following holds true:

Steady State

A transition matrix P is called *regular* if P^n has only strictly positive entries for some integer n . A *steady state* vector $\mathbf{q}$ of a transition matrix P is a probability vector that satisfies the equation $\mathbf{q}P = \mathbf{q}$. Notice that this implies $P'\mathbf{q}' = \mathbf{q}'$. For an eigenvalue of $\lambda = 1$ this also implies $P'\mathbf{q}' = \lambda\mathbf{q}'$ or $(P' - \lambda I)\mathbf{q}' = \mathbf{0}$. If P is a regular transition matrix the following holds true:

- 1 is an (left) eigenvalue of P or an eigenvalue of P'

Steady State

A transition matrix P is called *regular* if P^n has only strictly positive entries for some integer n . A *steady state* vector $\mathbf{q}$ of a transition matrix P is a probability vector that satisfies the equation $\mathbf{q}P = \mathbf{q}$. Notice that this implies $P'\mathbf{q}' = \mathbf{q}'$. For an eigenvalue of $\lambda = 1$ this also implies $P'\mathbf{q}' = \lambda\mathbf{q}'$ or $(P' - \lambda I)\mathbf{q}' = \mathbf{0}$. If P is a regular transition matrix the following holds true:

- 1 is an (left) eigenvalue of P or an eigenvalue of P'
- There is a unique probability vector $\mathbf{q}$, which is a (left) eigenvector associated with the eigenvalue 1 and $\mathbf{q}'$ is an eigenvector of P' .

Steady State

A transition matrix P is called *regular* if P^n has only strictly positive entries for some integer n . A *steady state* vector $\mathbf{q}$ of a transition matrix P is a probability vector that satisfies the equation $\mathbf{q}P = \mathbf{q}$. Notice that this implies $P'\mathbf{q}' = \mathbf{q}'$. For an eigenvalue of $\lambda = 1$ this also implies $P'\mathbf{q}' = \lambda\mathbf{q}'$ or $(P' - \lambda I)\mathbf{q}' = \mathbf{0}$. If P is a regular transition matrix the following holds true:

- 1 is an (left) eigenvalue of P or an eigenvalue of P'
- There is a unique probability vector $\mathbf{q}$, which is a (left) eigenvector associated with the eigenvalue 1 and $\mathbf{q}'$ is an eigenvector of P' .
- $\mathbf{x}P^n \rightarrow \mathbf{q}$ as $n \rightarrow \infty$ In the long-run this system converges to the steady state probability vector

Aperiodic, Communicate, Ergodic

- Two states communicate with each other if there is some positive probability of reaching one state from the other with a finite number of steps.

Aperiodic, Communicate, Ergodic

- A Markov Chain is aperiodic if for all states there is a positive probability of remaining in the same state in one step. A state can always return to the same state with some positive probability in the next period.

Aperiodic, Communicate, Ergodic

- A Markov Chain is aperiodic if for all states there is a positive probability of remaining in the same state in one step. A state can always return to the same state with some positive probability in the next period.
- A Markov Chain is Ergodic the chain has a unique steady state and converges to this steady state from all initial states.

Aperiodic, Communicate, Ergodic

- A Markov Chain is aperiodic if for all states there is a positive probability of remaining in the same state in one step. A state can always return to the same state with some positive probability in the next period.
- A Markov Chain is Ergodic the chain has a unique steady state and converges to this steady state from all initial states.
 - A Markov Chain is Ergodic if the process is aperiodic and all states communicate.

Steady State: Example

What is the probability it will be cloudy as $n \rightarrow \infty$? Our transition probability matrix is

$$\begin{pmatrix} .8 & .1 & .1 \\ .3 & .2 & .5 \\ .2 & .6 & .2 \end{pmatrix}$$

The matrix P is regular since P only has strictly positive entries. Then to find the long-run probabilities we must find the vector $\mathbf{q}$. First we use the fact that when $\lambda = 1$ it is the case that $\mathbf{q}P = \lambda\mathbf{q} \implies P'\mathbf{q}' = \lambda\mathbf{q}'$. Therefore, $\mathbf{q}'$ is the eigenvector of $\mathbf{P}'$ that corresponds to the eigenvalue $\lambda = 1$. So we must find this eigenvector.

Steady State: Example

$$(P' - \lambda I)\mathbf{q}' = \begin{pmatrix} -.2 & .3 & .2 \\ .1 & -.8 & .6 \\ .1 & .5 & -.8 \end{pmatrix} \mathbf{q}' = 0 \implies \left(\begin{array}{ccc|c} -.2 & .3 & .2 & 0 \\ .1 & -.8 & .6 & 0 \\ .1 & .5 & -.8 & 0 \end{array} \right)$$
$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{34}{13} & 0 \\ 0 & 1 & -\frac{14}{13} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \implies \begin{aligned} q_1 &= \frac{34}{13} q_3 \\ q_2 &= \frac{14}{13} q_3 \\ q_3 &= q_3 \end{aligned}$$

Therefore $\mathbf{q} = (\frac{34}{13}, \frac{14}{13}, 1)q_3$, where q_3 can be any value. Since the elements of $\mathbf{q}$ must sum to 1 we chose q_3 accordingly

$$\frac{34}{13} + \frac{14}{13} + \frac{13}{13} = \frac{61}{13} \implies q_3 = \frac{13}{61} \implies \mathbf{q} = (\frac{34}{61}, \frac{14}{61}, \frac{13}{61}).$$

Ergodic Failure

Consider the case, in which $\mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Then, chain just fluctuates between state 1 and state 2. So the states communicate but the chain is not aperiodic. It takes two periods to for a state to return. The unique steady state in this

case is $\begin{pmatrix} .5 \\ .5 \end{pmatrix}$, but the chain only converges to this state if

$$\mathbf{x}_0 = \begin{pmatrix} .5 \\ .5 \end{pmatrix}$$